

# Neural Architectures for Geospatial Intelligence

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Shang-Ling Hsu

CSCI-587: Geospatial Data Analysis

Spring 2026

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# Background Survey

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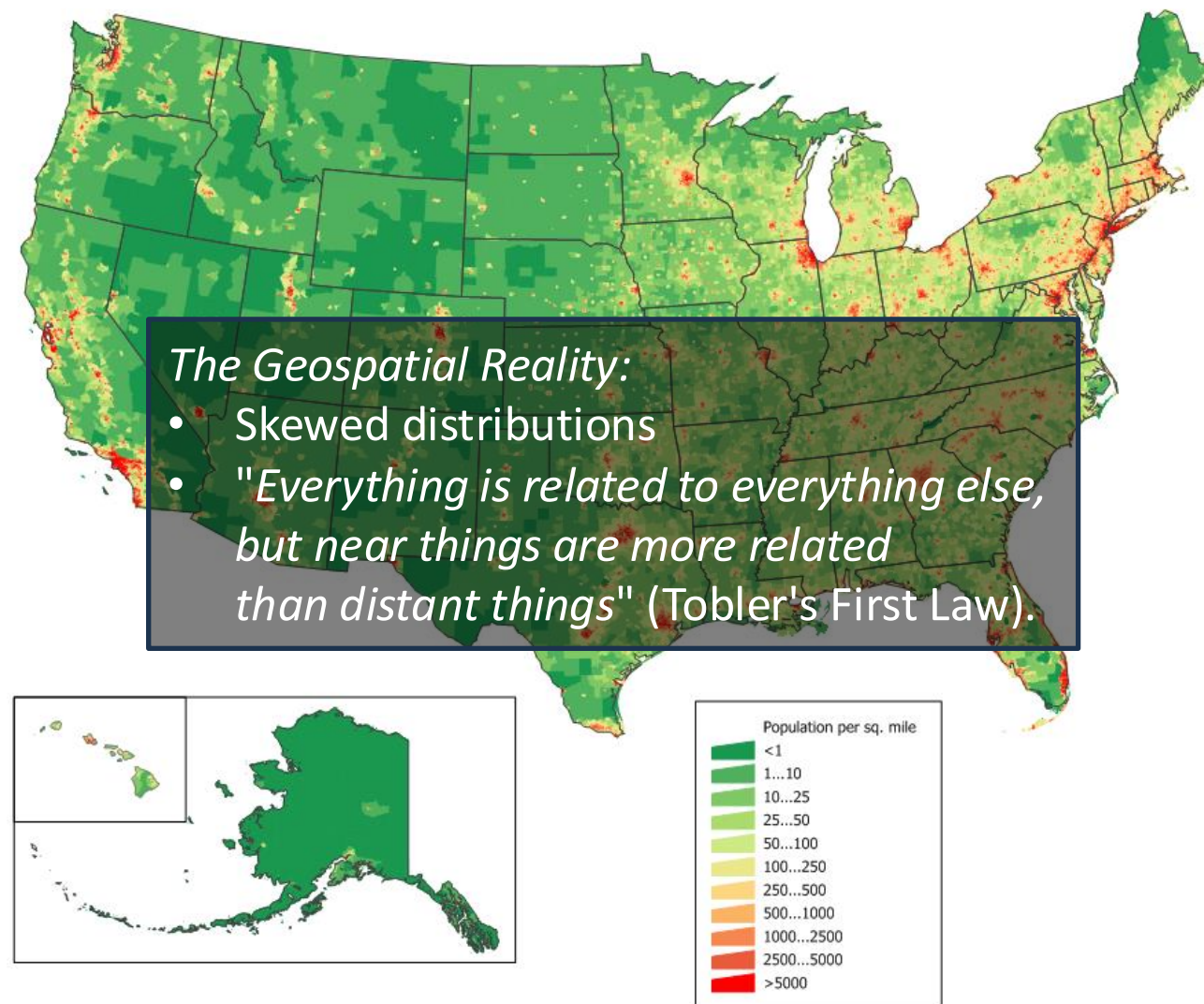


# The "Independent Data" Fallacy

Standard ML assumes data points are *independent and identically distributed (i.i.d.)*.

*The Geospatial Reality:*

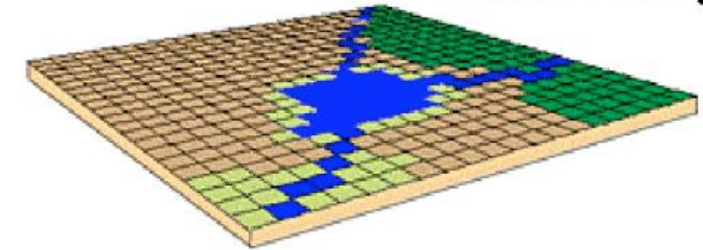
- Skewed distributions
- "Everything is related to everything else, but near things are more related than distant things" (Tobler's First Law).



# Data Modalities

- **Grids:** Raster data
- **Sequences:** GPS trajectories, Time-series.
- **Graphs:** Road networks, Sensor webs.
- **Images:**
  - Satellite and aerial photography
  - Geo-tagged photos and videos
  - Street view images
- **Others:**
  - Text reviews
  - Opening hours
  - ... and more!

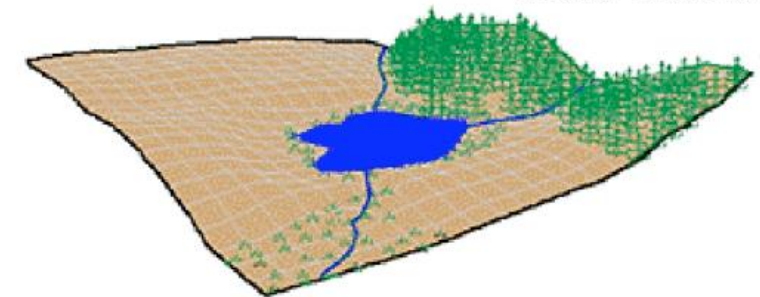
**Raster / Image**



**Vector**



**Real World**



| Lecture |                                                                                                                     |
|---------|---------------------------------------------------------------------------------------------------------------------|
| 20      | <u>Semi-Supervised Classification with Graph Convolutional Networks</u>                                             |
| 20      | <u>Diffusion Convolutional Recurrent Neural Network; Data-driven Traffic Forecasting</u>                            |
| 20      | <u>Learning Dynamic Graphs from All Contextual Information for Accurate Point-of-Interest Visit Forecasting</u>     |
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| 25      | <u>MULTI-SCALE REPRESENTATION LEARNING FOR SPATIAL FEATURE DISTRIBUTIONS USING GRID CELLS</u>                       |
| 25      | <u>Poly2Vec</u>                                                                                                     |
| 26      | <u>Hierarchical Transformer with Spatio-temporal Context Aggregation for Next Point-of-Interest Recommendation</u>  |
| 26      | <u>TrajGPT; Controlled Synthetic Trajectory Generation Using a Multitask Transformer-Based Spatiotemporal Model</u> |

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- Introduction
- Neural Architectures
- Advanced Training Paradigms
- Conclusion



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# Convolutional Neural Networks (CNNs)



<https://www.maptiler.com/maps/satellite/>



<https://www.istockphoto.com/photos/new-york-city-street-view>



<https://transportation.engineering.gwu.edu/traffic-flow-theory-and-driver-behavioral-modeling>



# Convolution\* in 1D

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- Input vector  $\mathbf{x}$ :

$$\mathbf{x} = [x_1, x_2, \dots, x_I]$$

- Output is weighted sum of neighbors:

$$z_i = \omega_1 x_{i-1} + \omega_2 x_i + \omega_3 x_{i+1}$$

- Convolutional **kernel** or **filter**:

$$\boldsymbol{\omega} = [\omega_1, \omega_2, \omega_3]^T$$


Kernel size = 3

\* Not really technically convolution



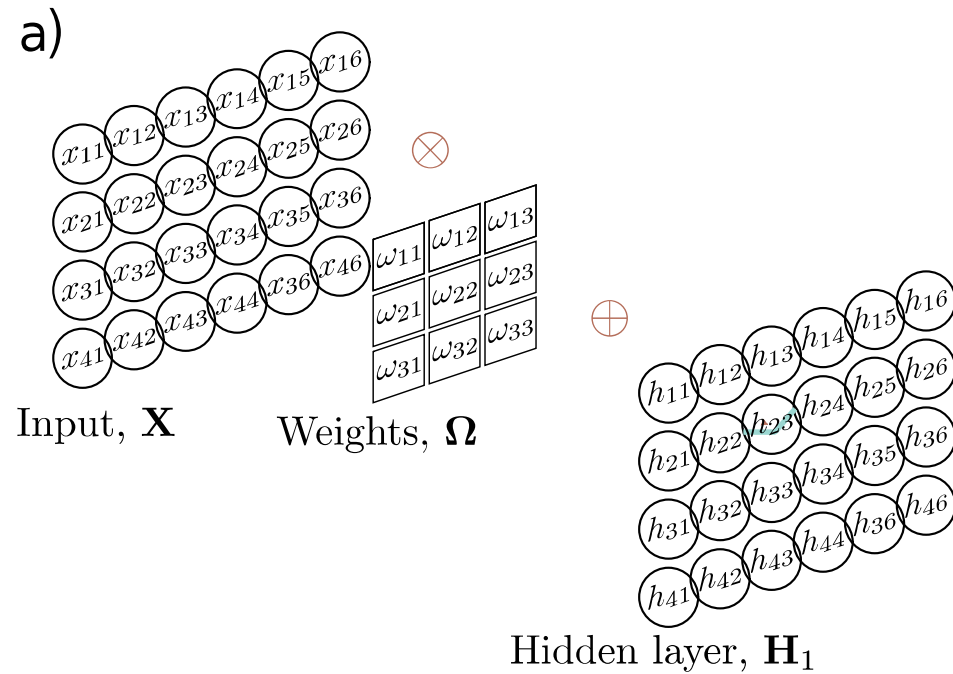
# Convolution in 2D

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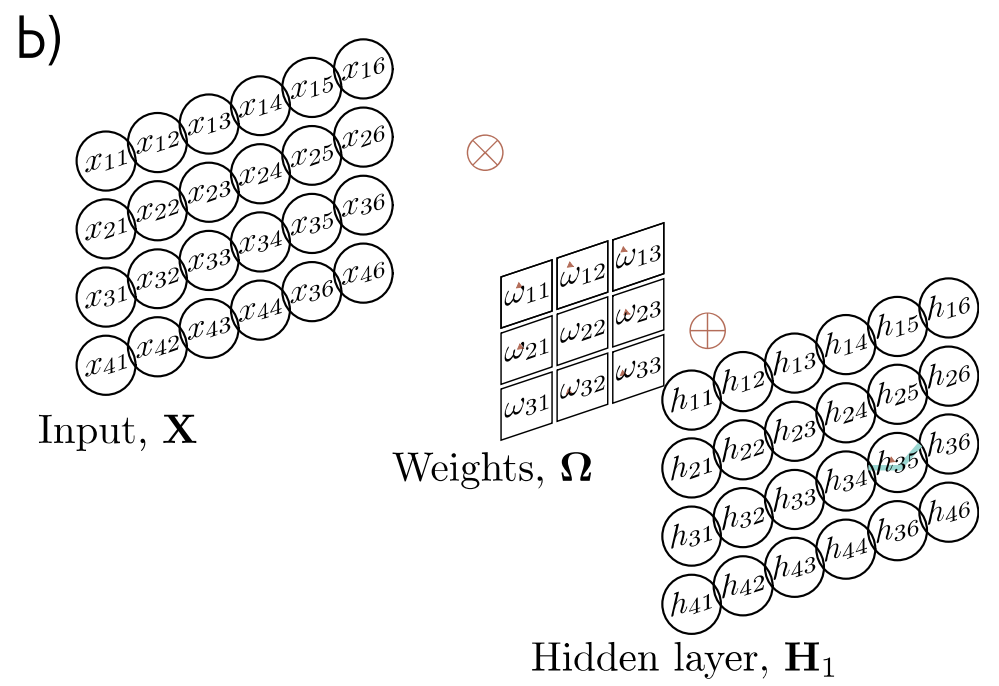
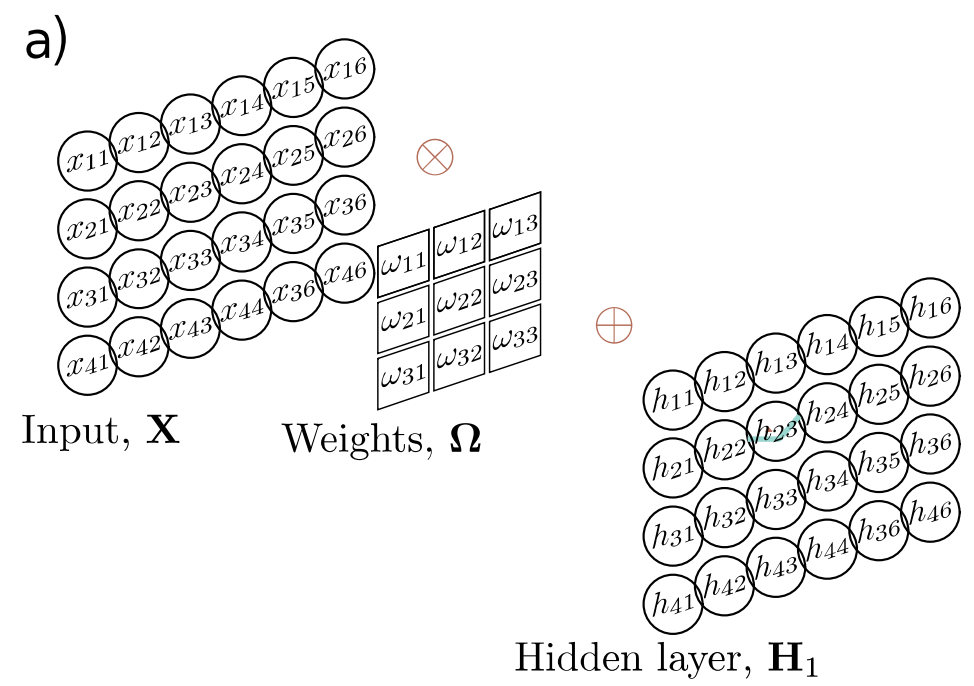
- Convolution in 2D
  - Weighted sum over a  $K \times K$  region
  - $K \times K$  weights
- Build into a convolutional layer by adding bias and passing through activation function

$$h_{i,j} = a \left[ \beta + \sum_{m=1}^3 \sum_{n=1}^3 \omega_{m,n} x_{i+m-2,j+n-2} \right]$$

# Convolution in 2D

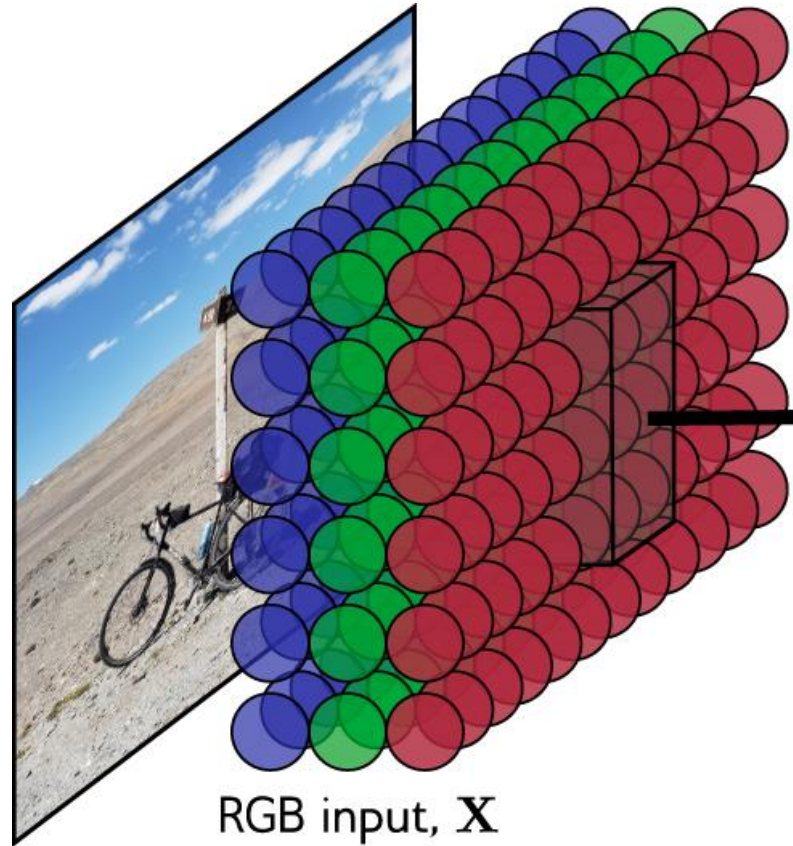


# Convolution in 2D

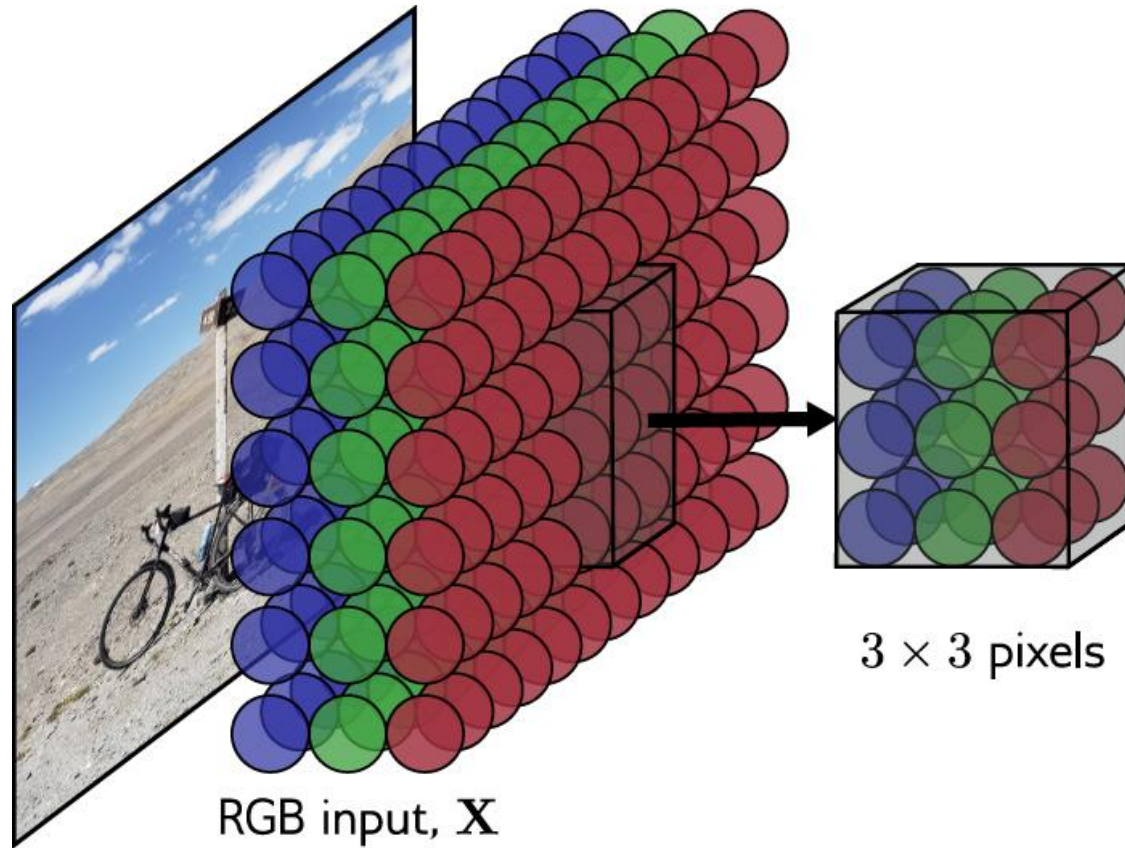




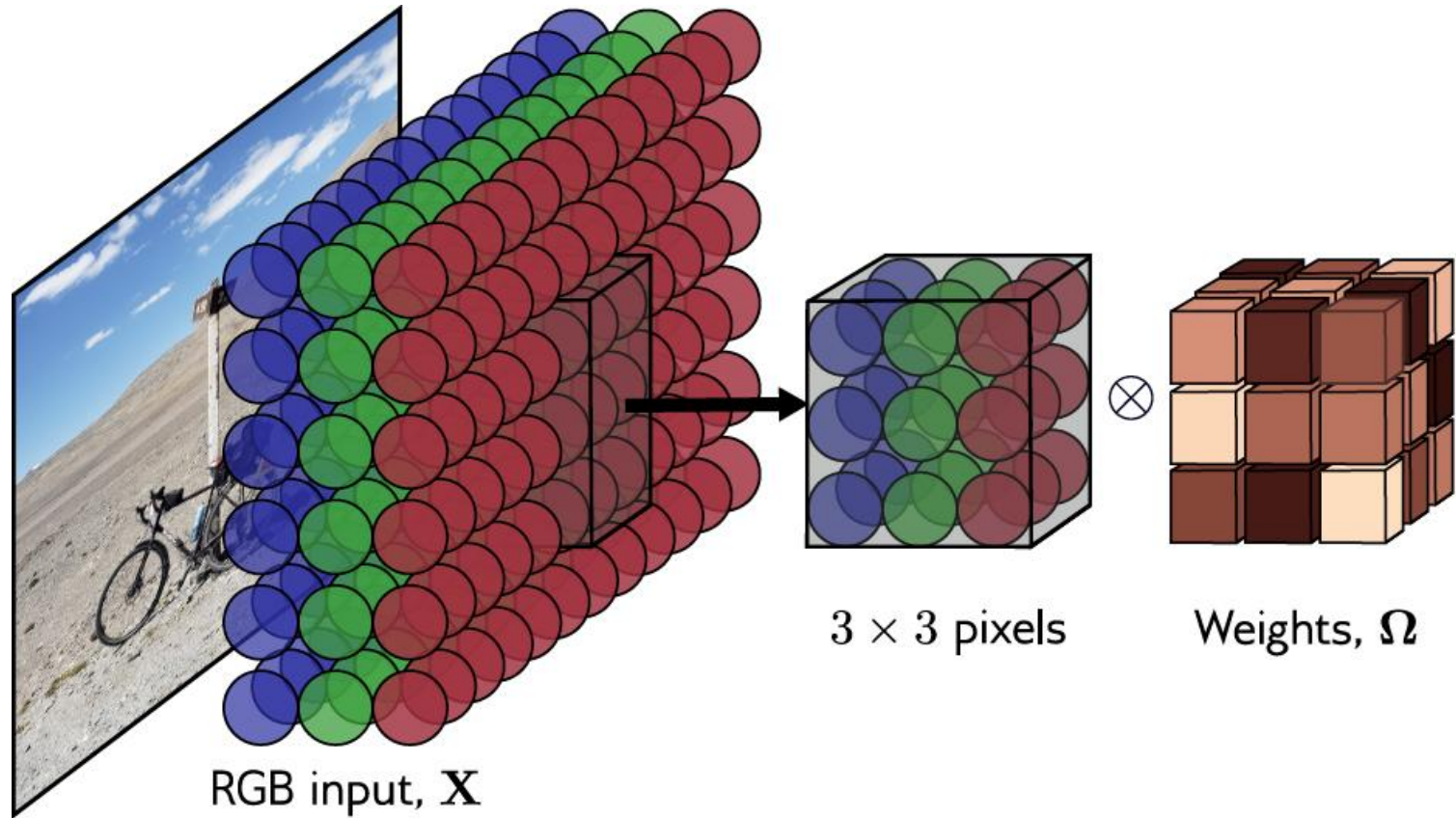
# Channels in 2D convolution



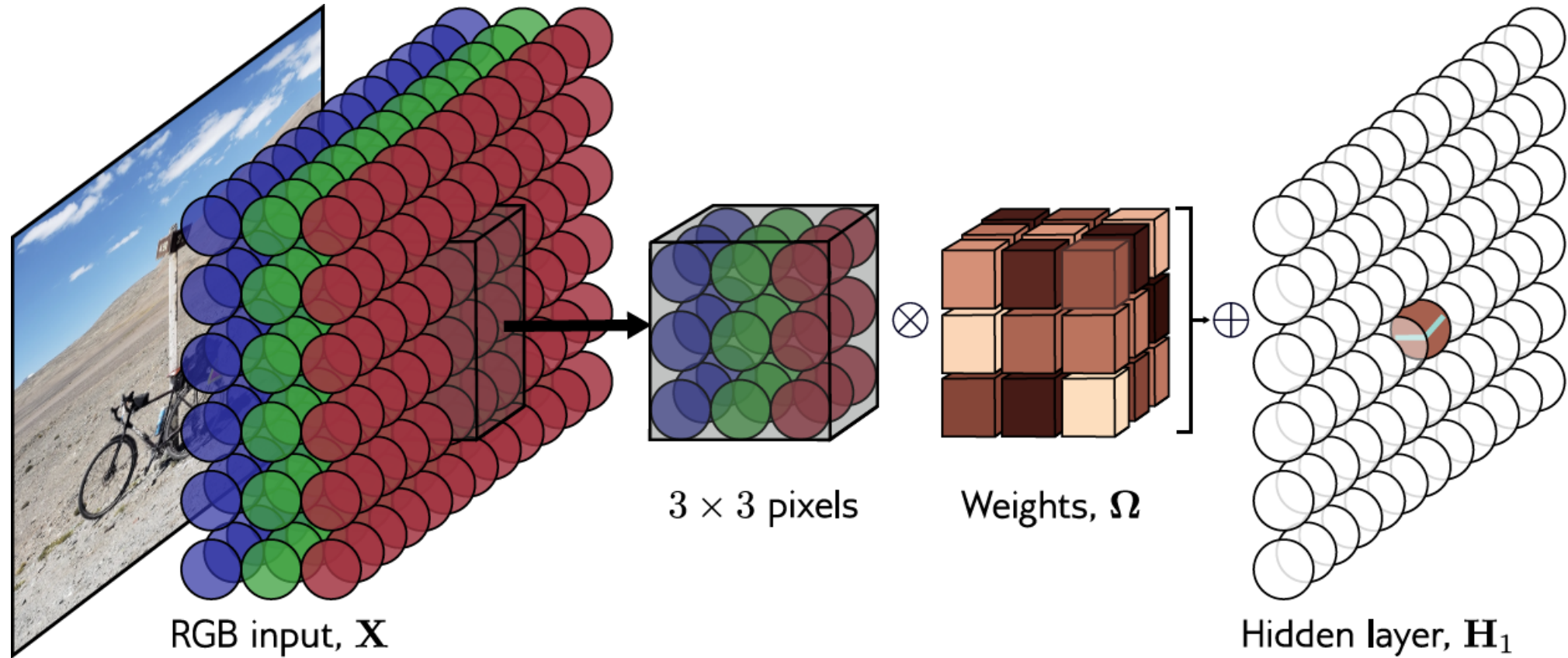
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# Channels in 2D convolution



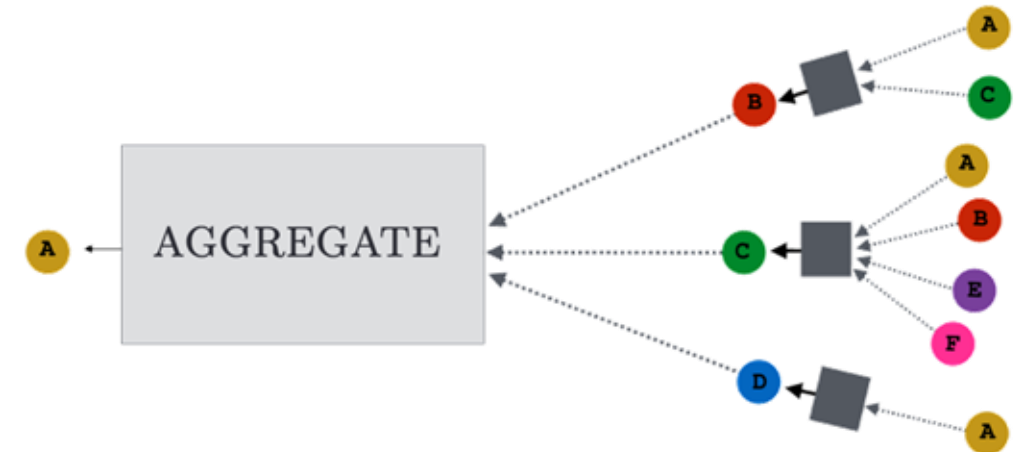
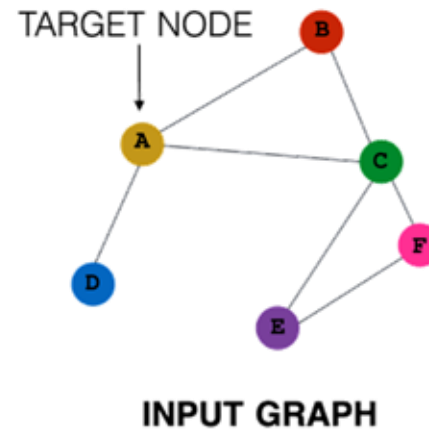
# Channels in 2D convolution





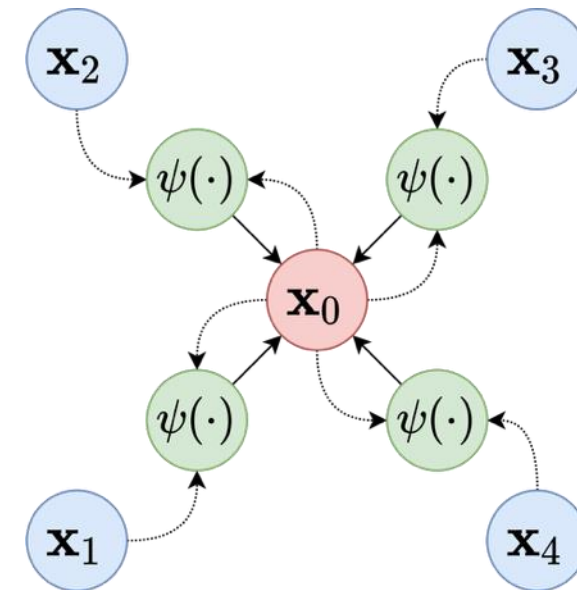
# Generalizing Convolutions to Graphs

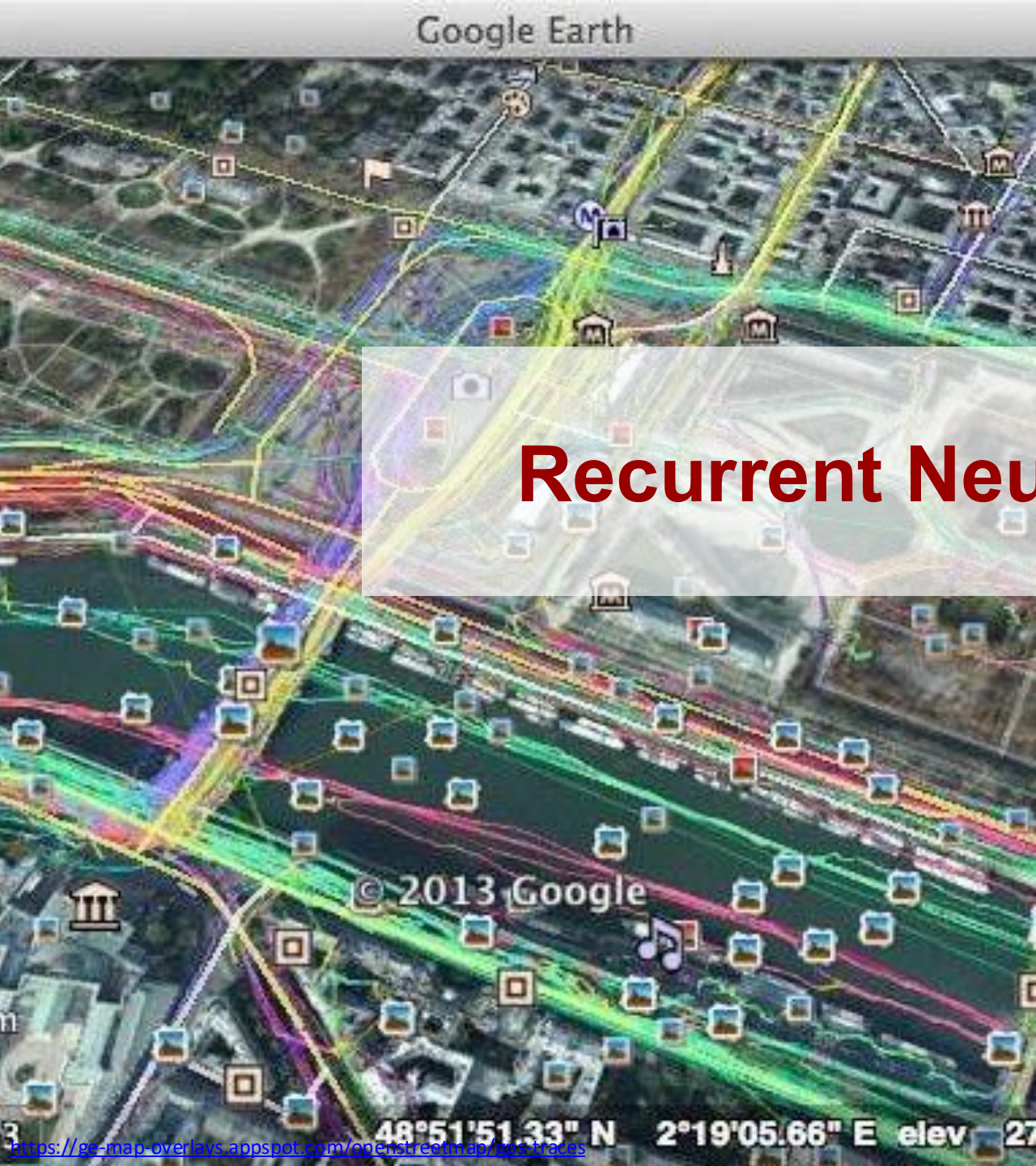
- Message passing



- Neural message passing (graph convolution)
  - Target node:  $x_0$
  - Neighbors:  $x_1, x_2, x_3, x_4$
  - Message function:  $\psi$

$$\mathbf{h}_u = \phi \left( \mathbf{x}_u, \bigoplus_{v \in N_u} \psi(\mathbf{x}_u, \mathbf{x}_v, \mathbf{e}_{uv}) \right)$$





## Recurrent Neural Networks (RNNs)

### Popular times <sup>?</sup>

MON

TUE

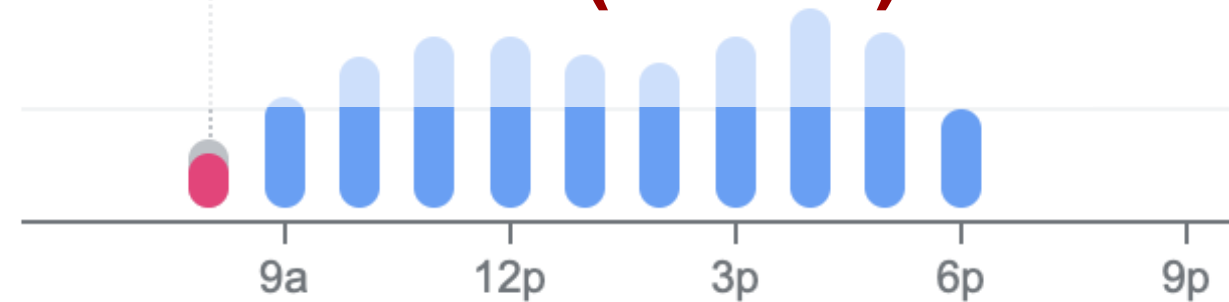
WED

THU

FRI

SAT

● **Live:** Not busy



People typically spend **25 min** here

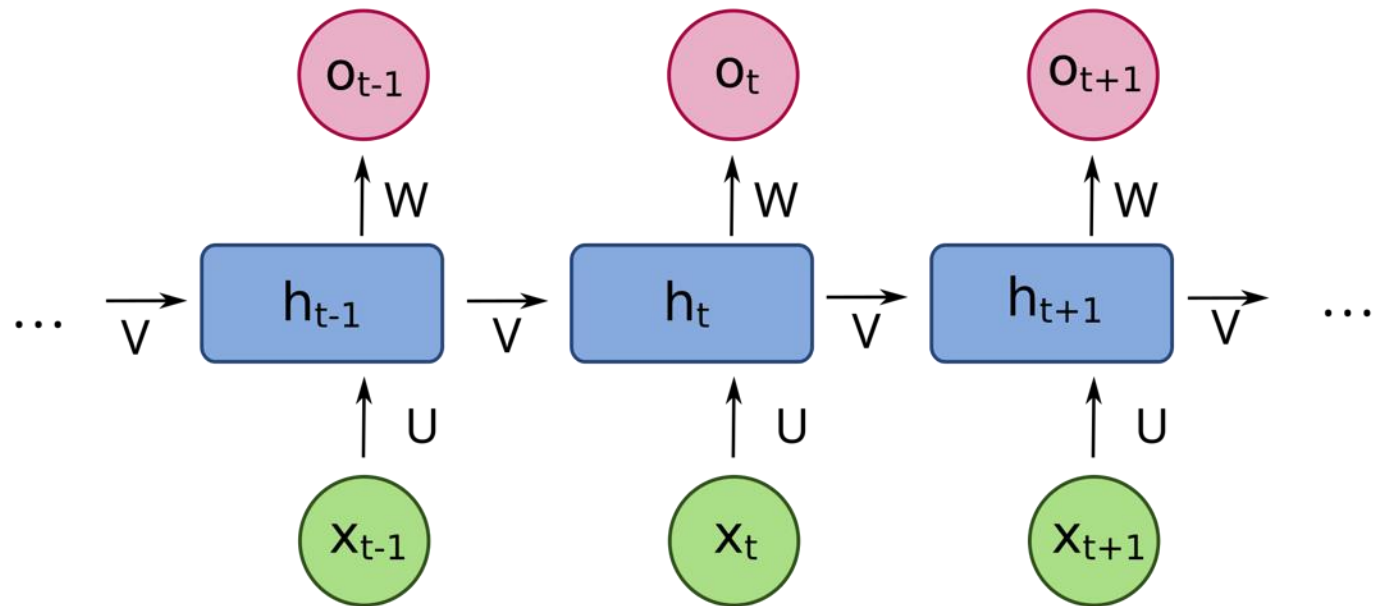


**Send to your phone**

# Standard RNNs

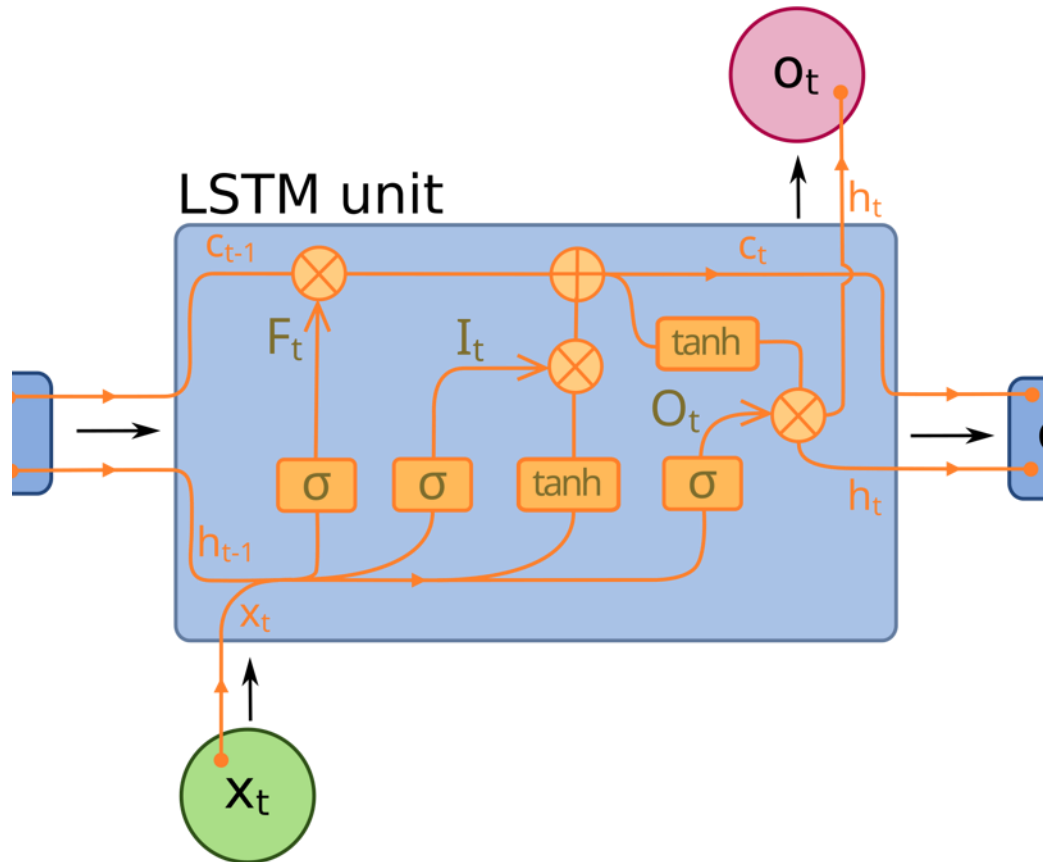
General form:

- $O_t$  : output vector
- $X_t$ : input vector
- $h_t$  : hidden vector (“memory”)
- $U, V, W$ : weights  
(neural network parameters)



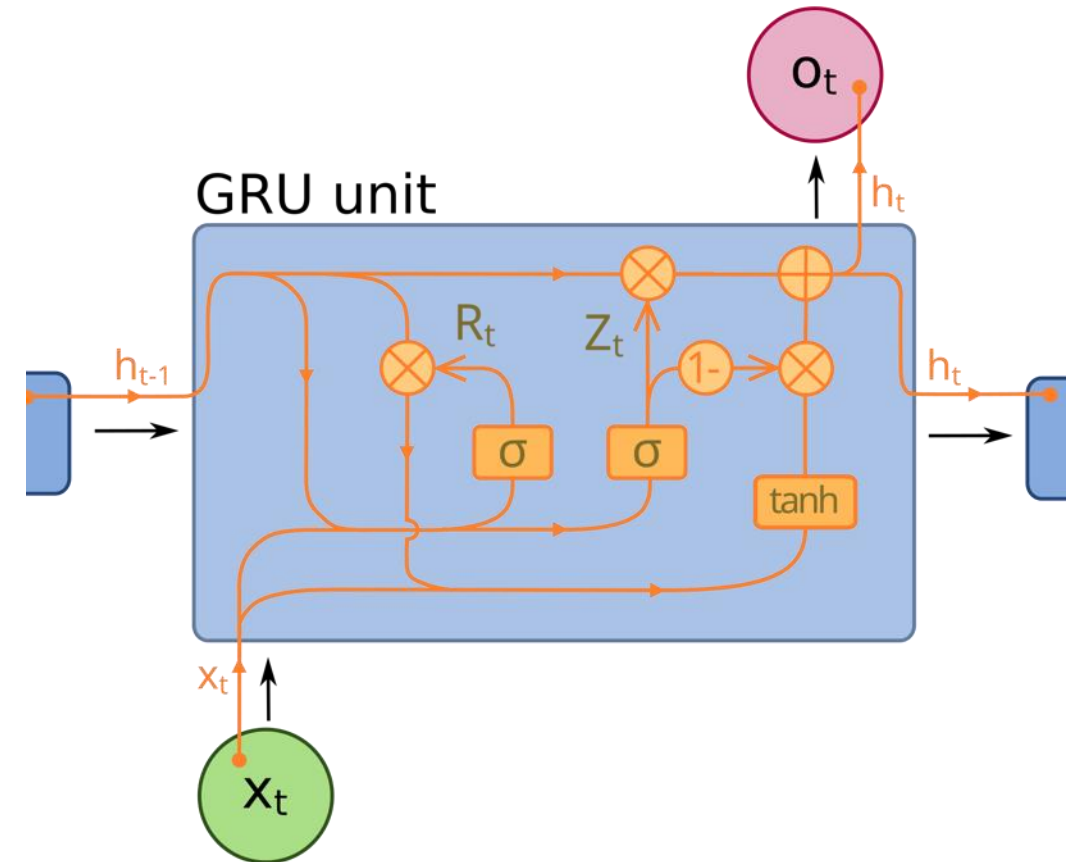


# RNN Variants



## Long Short-Term Memory (LSTM)

To mitigate the vanishing gradient problem.



## Gated Recurrent Unit (GRU)

Fewer parameters than LSTM.



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# Transformers

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*Ashish Vaswani 🦁, Noam Shazeer, Niki Parmar 🦁, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, Illia Polosukhin. “Attention Is All You Need.” NeurIPS 2017.*

# Transformer

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Word embeddings



The  —

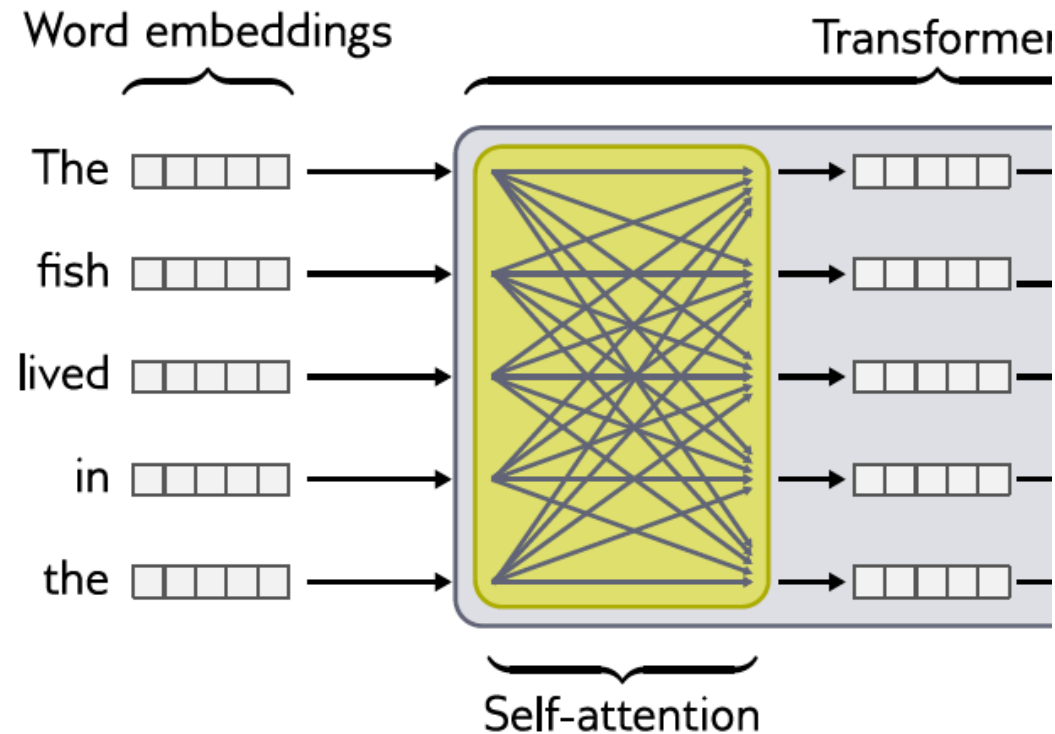
fish  —

lived  —

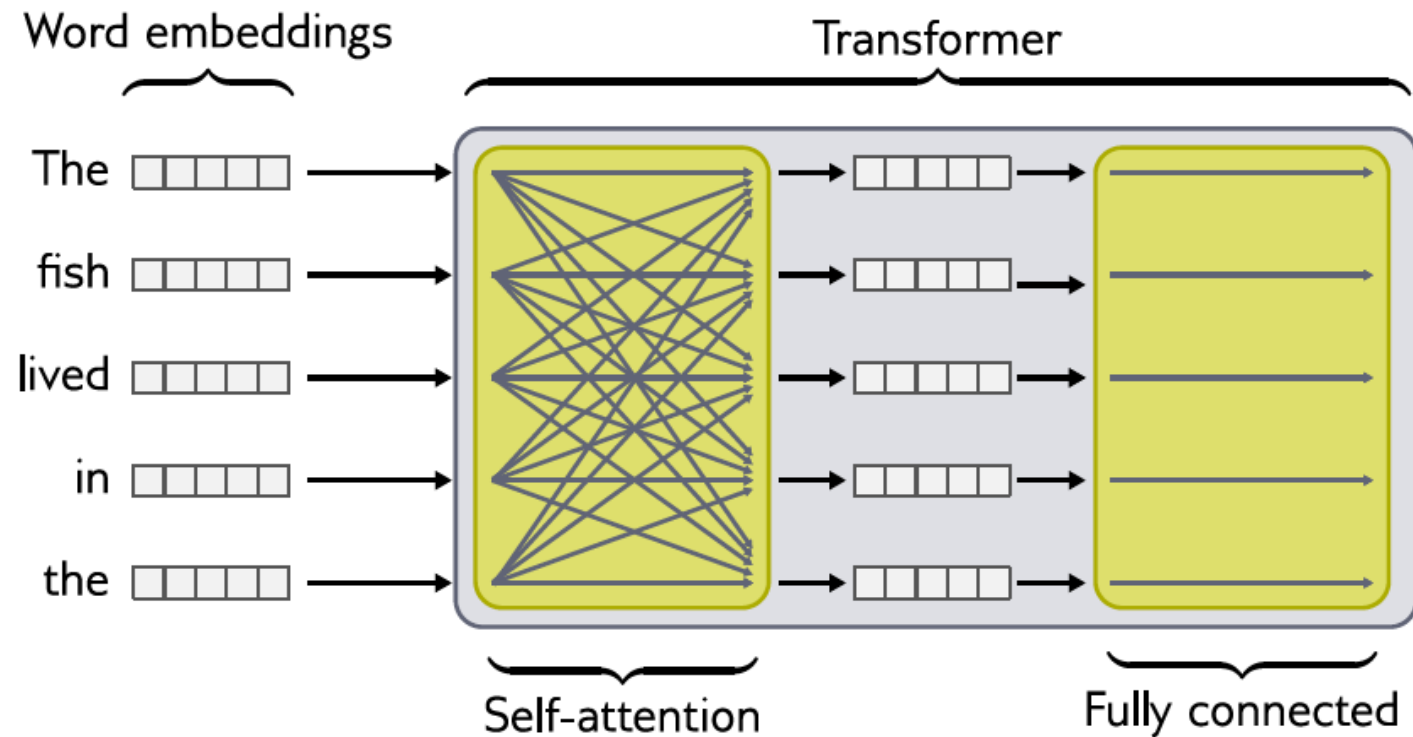
in  —

the  —

# Transformer

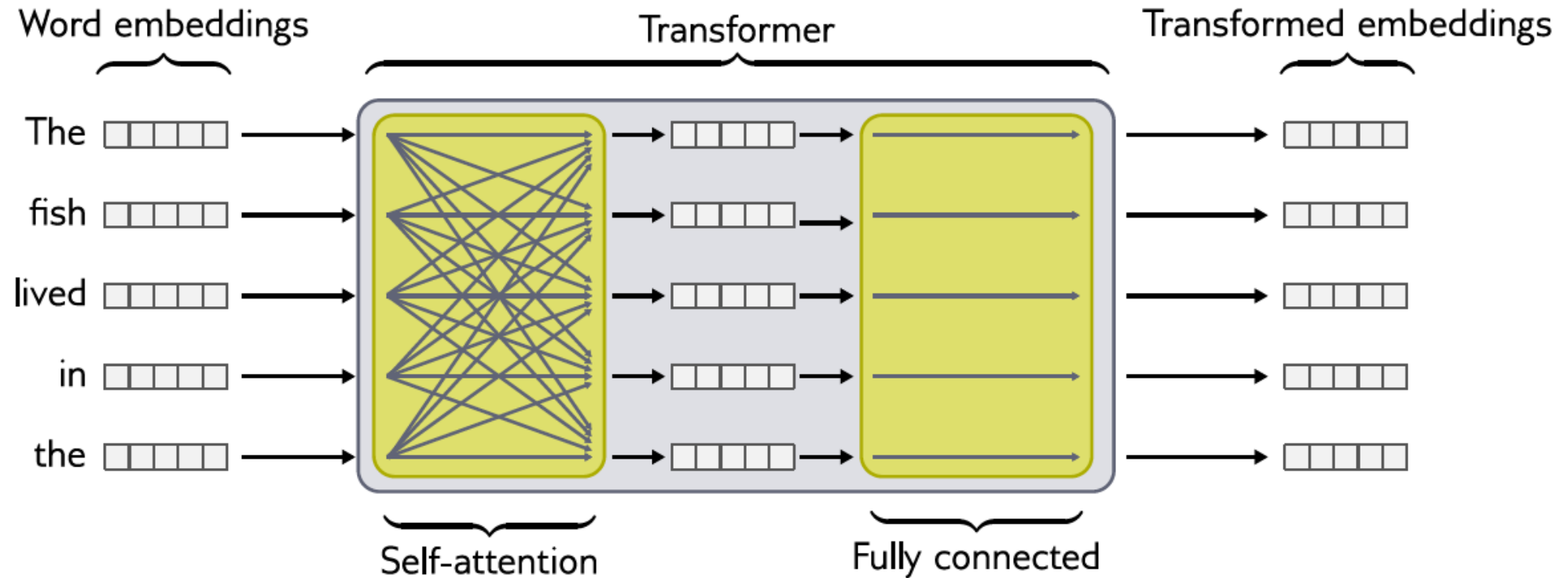


# Transformer

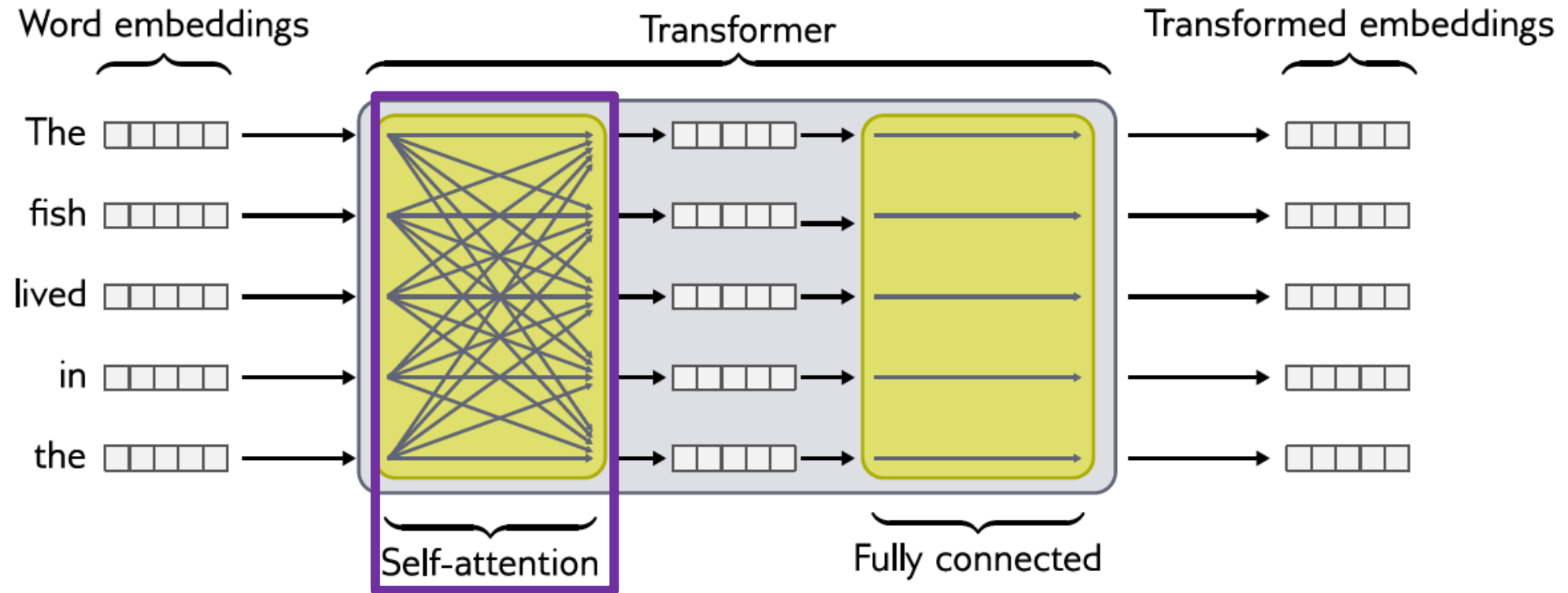




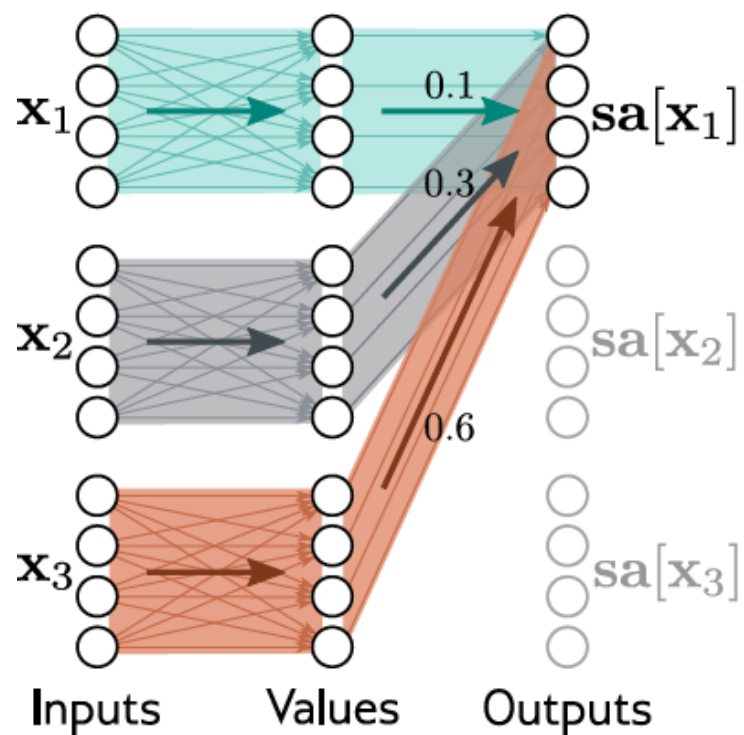
# Transformer



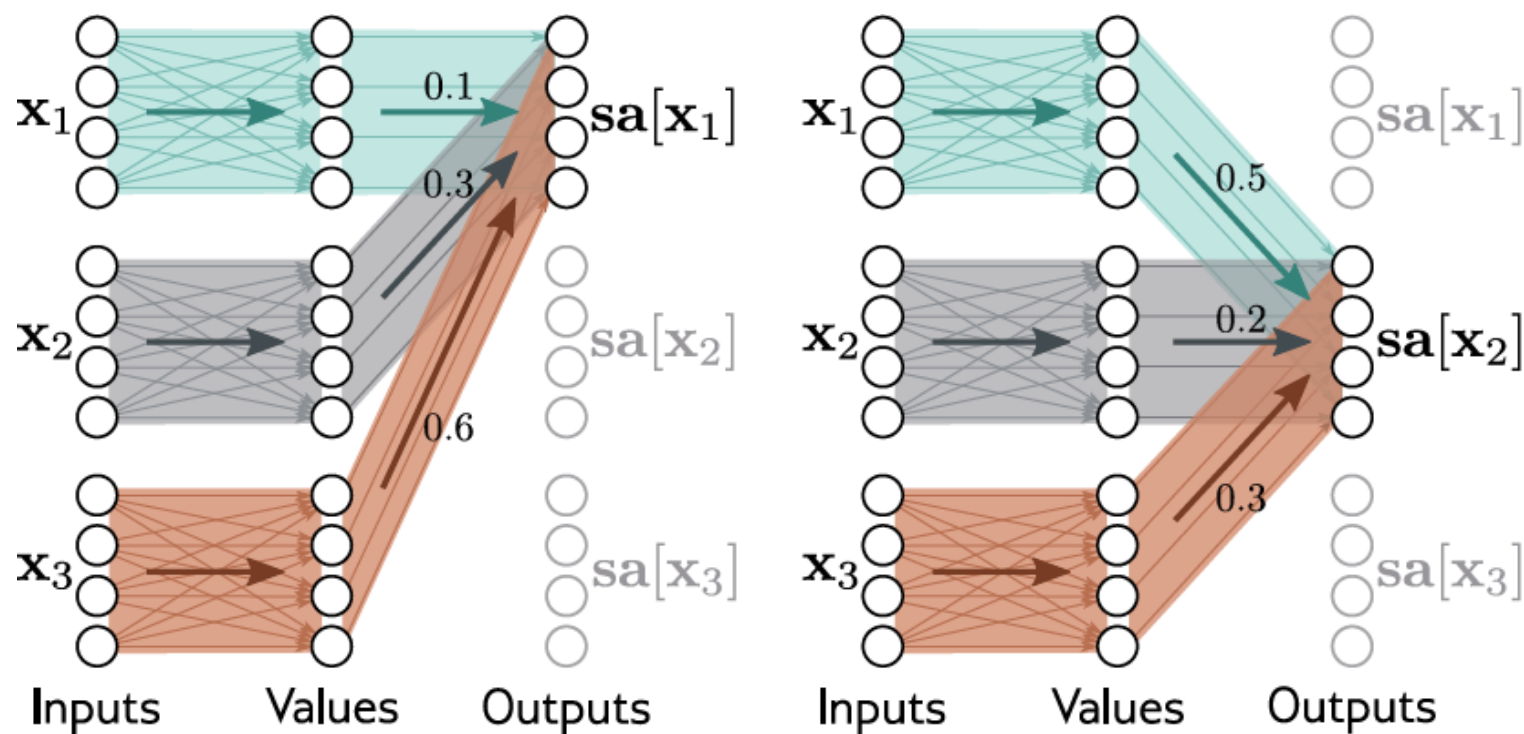
# Transformer



# Self-Attention

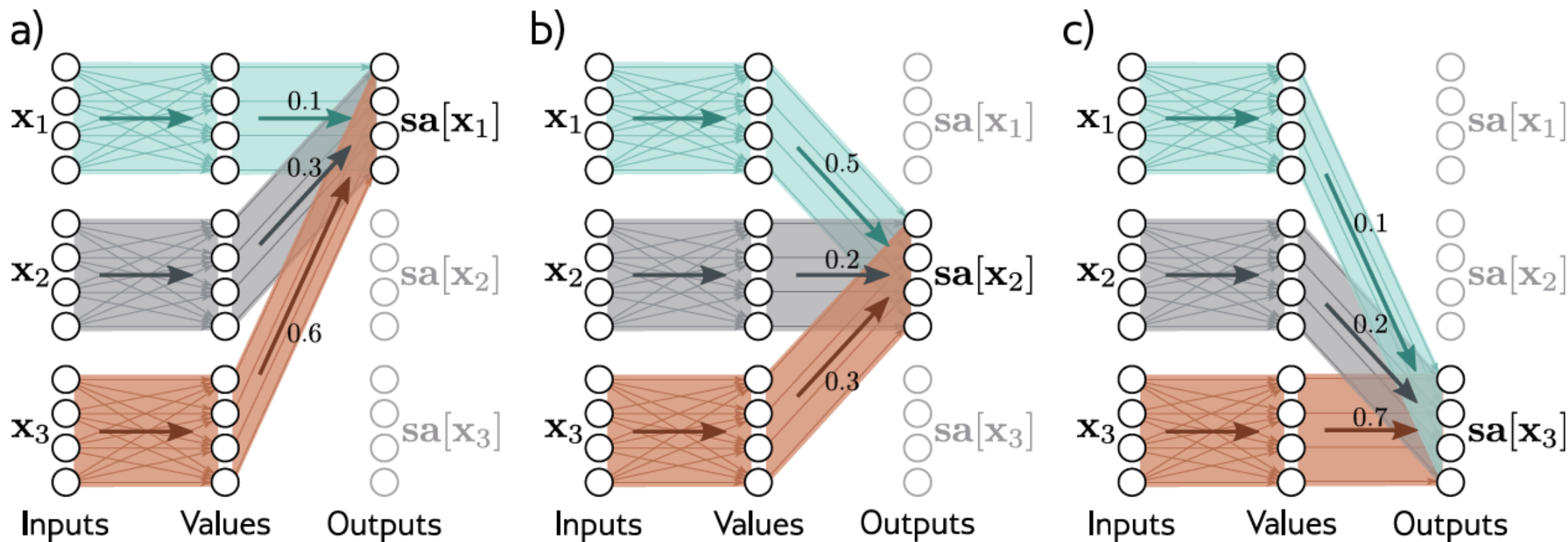


# Self-Attention





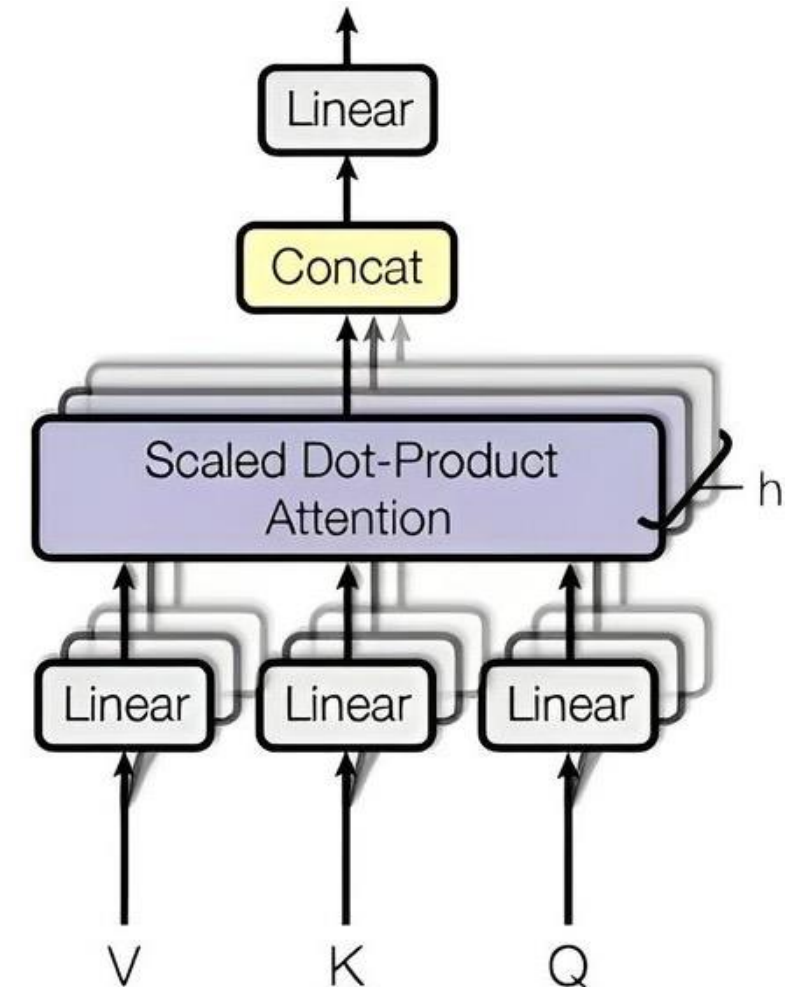
# Self-Attention



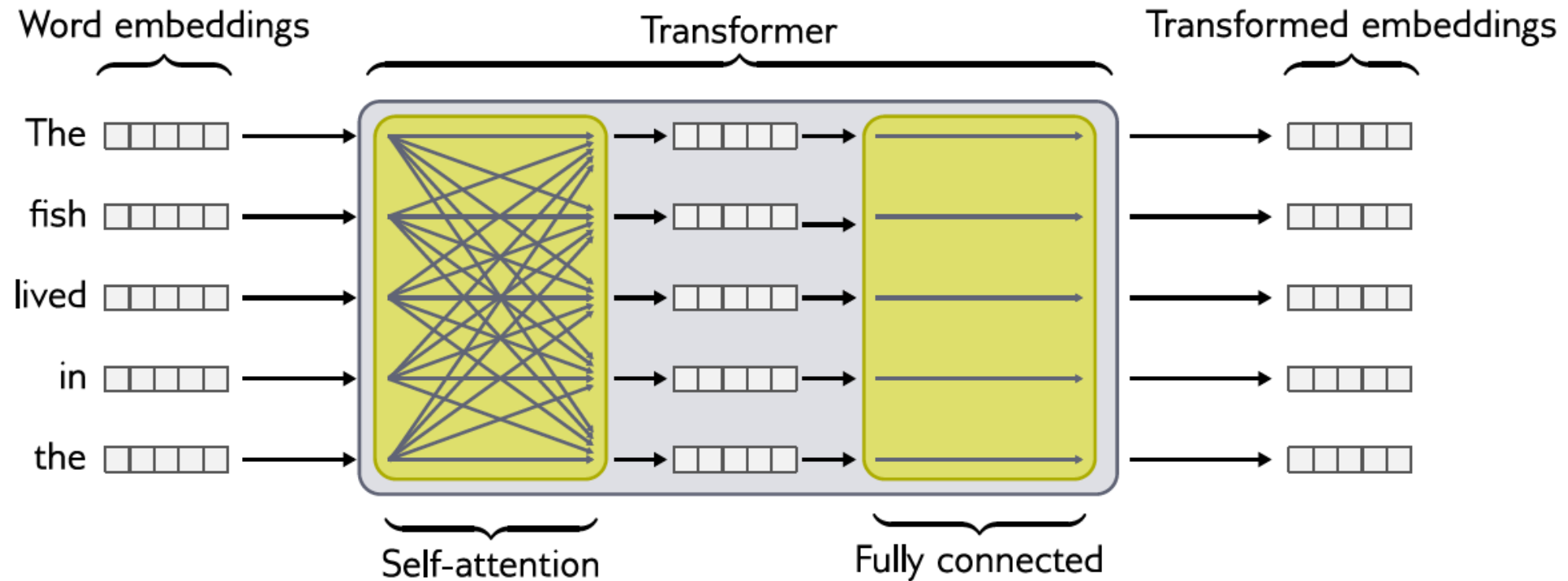
# Multi-Head Self-Attention

Compute self-attention multiple times with

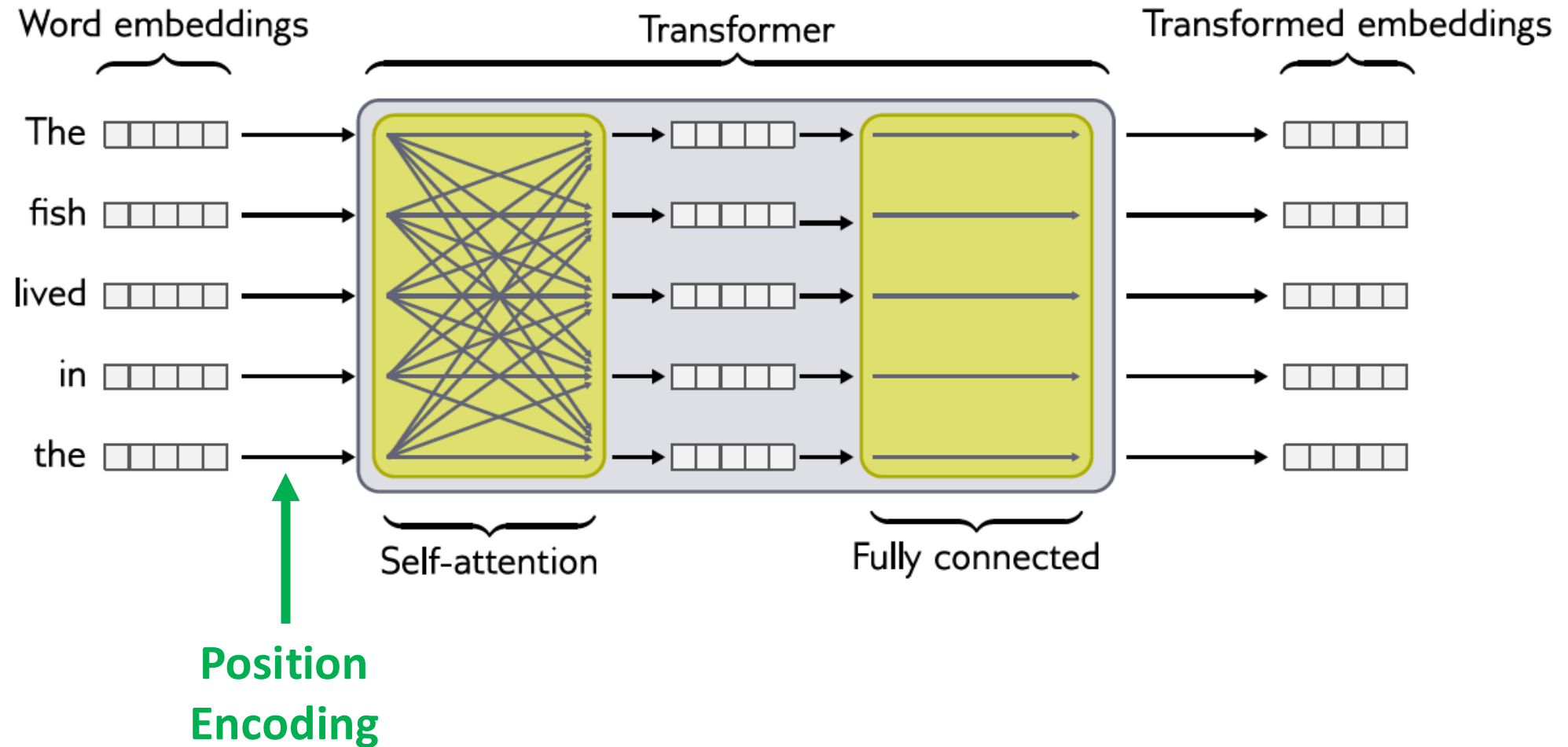
- the same input, but
  - a different set of model parameters.
- Capturing various types of information.



# Transformer



# Transformer



# Positional Encoding

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Self-attention is equivariant to permuting word order

But word order is important in language:

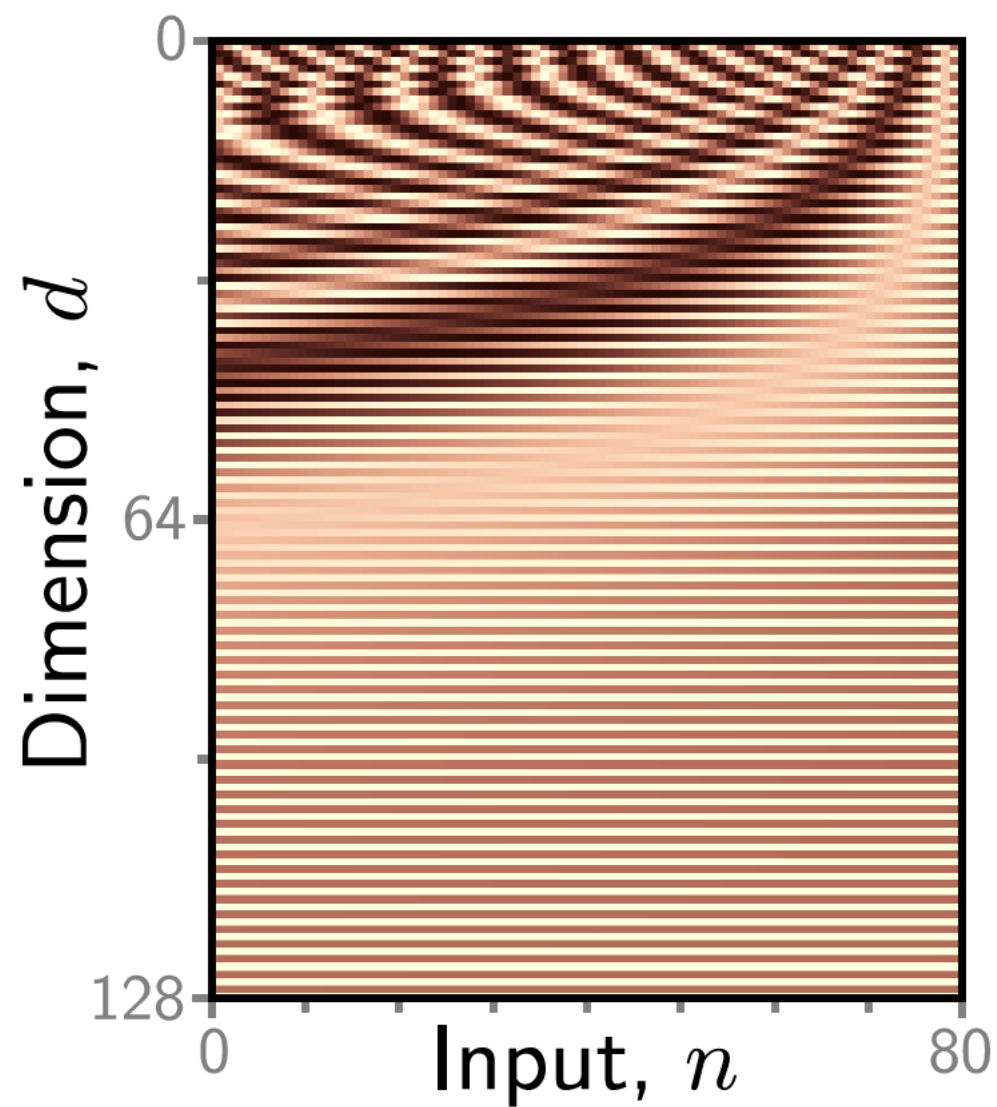
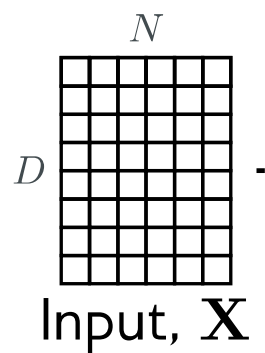
The man ate the fish

vs.

The fish ate the man

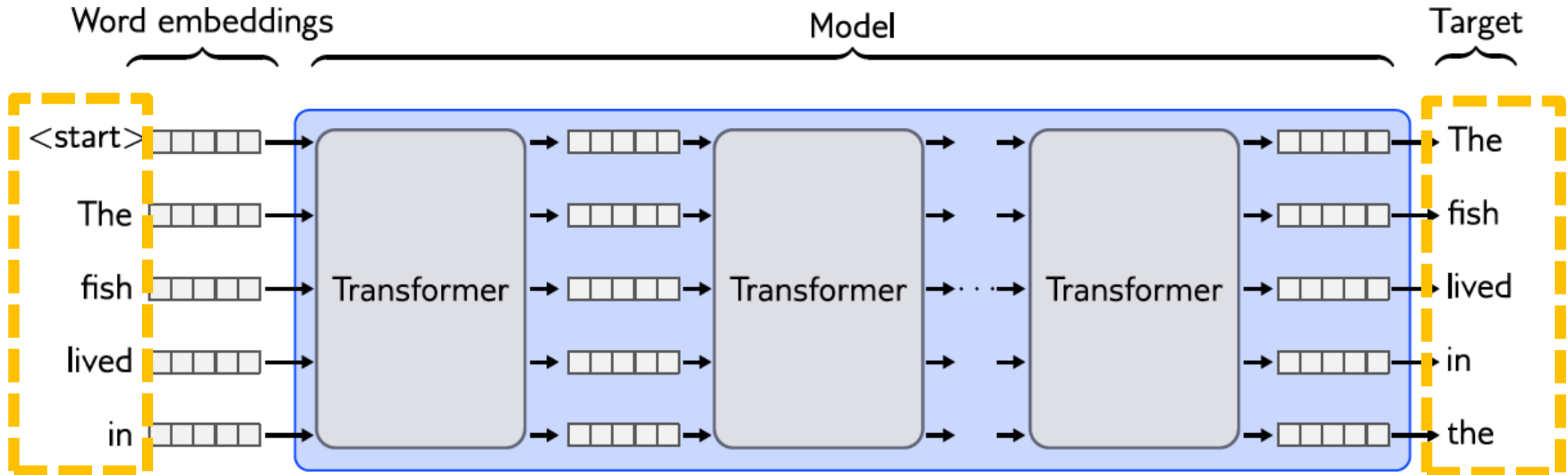


# Positional Encoding



# Generative Pre-trained Transformer (GPT)

- Predicting all next words simultaneously:



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# Variational Autoencoders (VAEs)

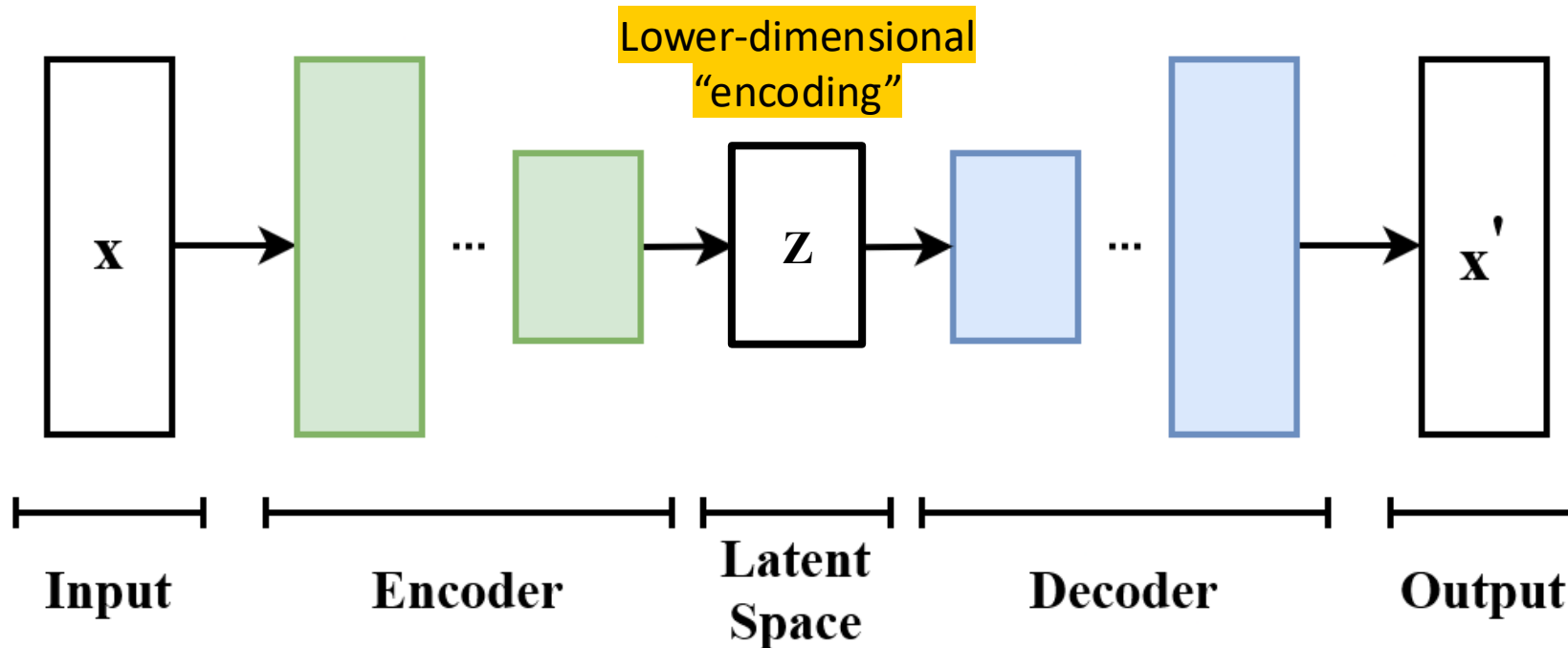
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*Diederik P. Kingma, Max Welling. "Auto-Encoding Variational Bayes." ICLR 2014.*



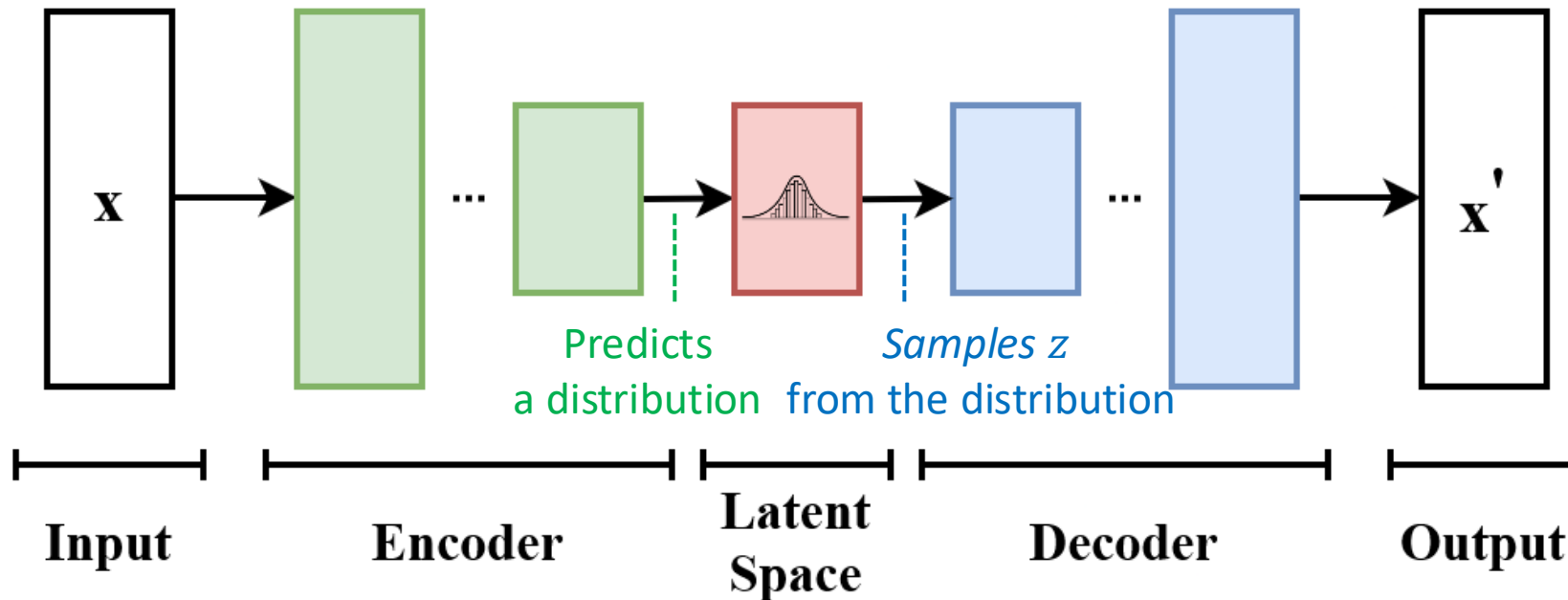
# Variational Autoencoders

- To learn efficient encodings for *compression* and *reconstruction*.
- Unseen  $z$ ?  $\rightarrow$  nonsense  $x'$  😞



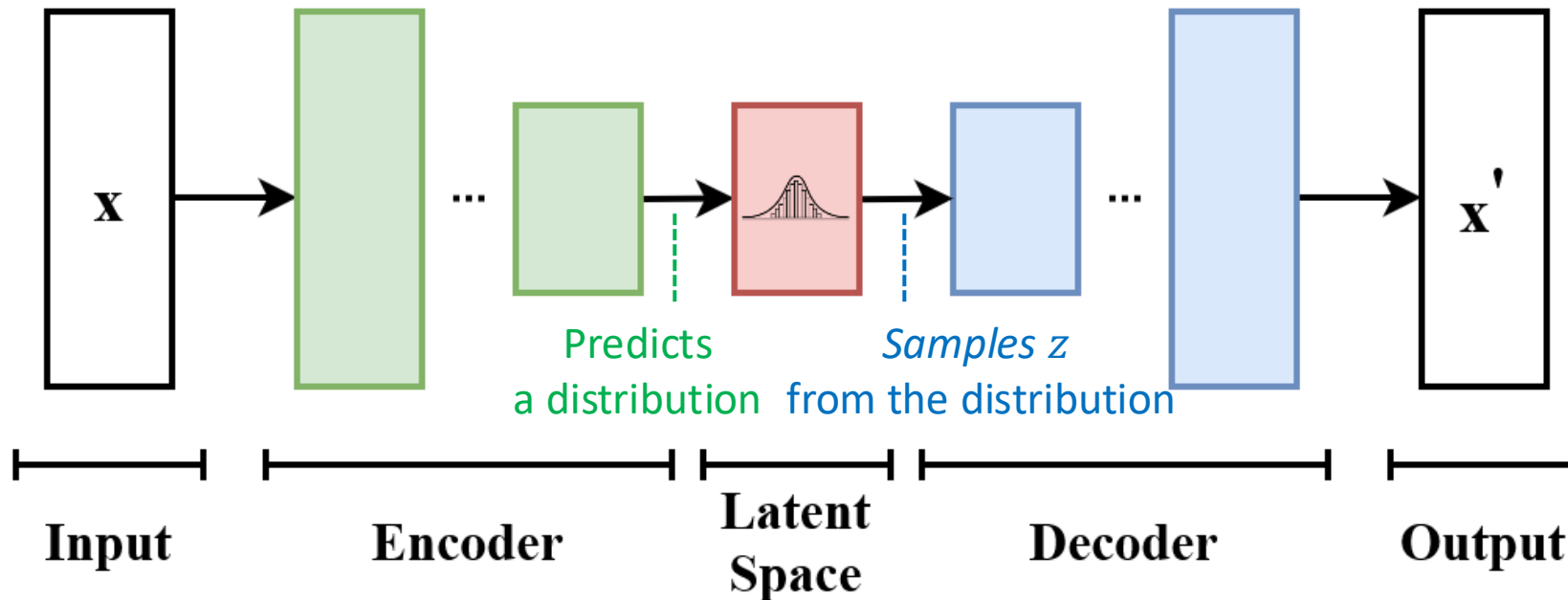
# Variational Autoencoders (VAEs)

- To learn to ✨ *generate* ✨ new data.  
→ To learn a smooth and continuous latent space.



# Variational Autoencoders (VAEs)

- To learn to ✨ *generate* ✨ new data.
  - To learn a smooth and continuous latent space.
  - How can we learn the model parameters?

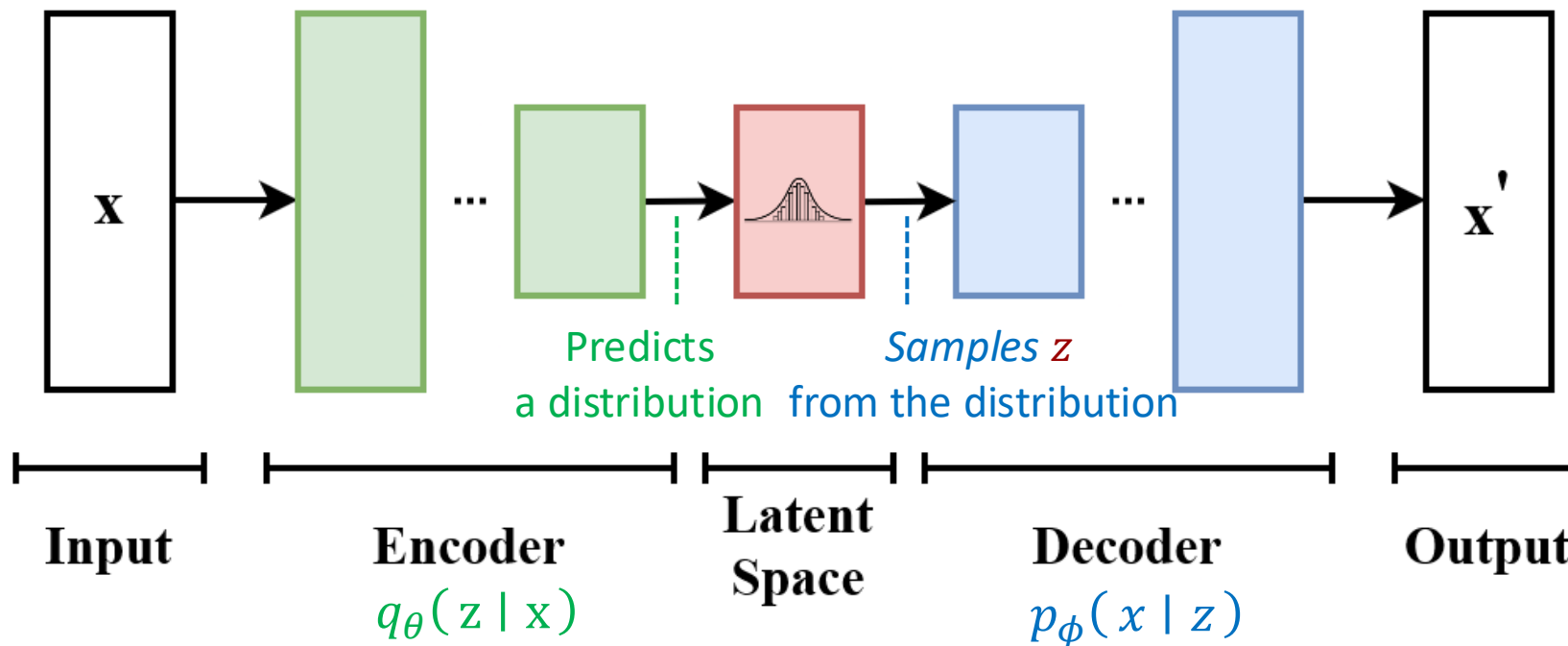




# Learning VAE Parameters $(\theta, \phi)$

Maximize its *evidence Lower bound*, “**ELBO**”: ← Proofs are in the optional reading.

$$L_{\theta, \phi}(x) := \mathbb{E}_{z \sim q_{\theta}(\cdot | x)} [\ln p_{\phi}(x | z)] - D_{KL}(q_{\theta}(\cdot | x) \parallel p_{\phi}(\cdot))$$



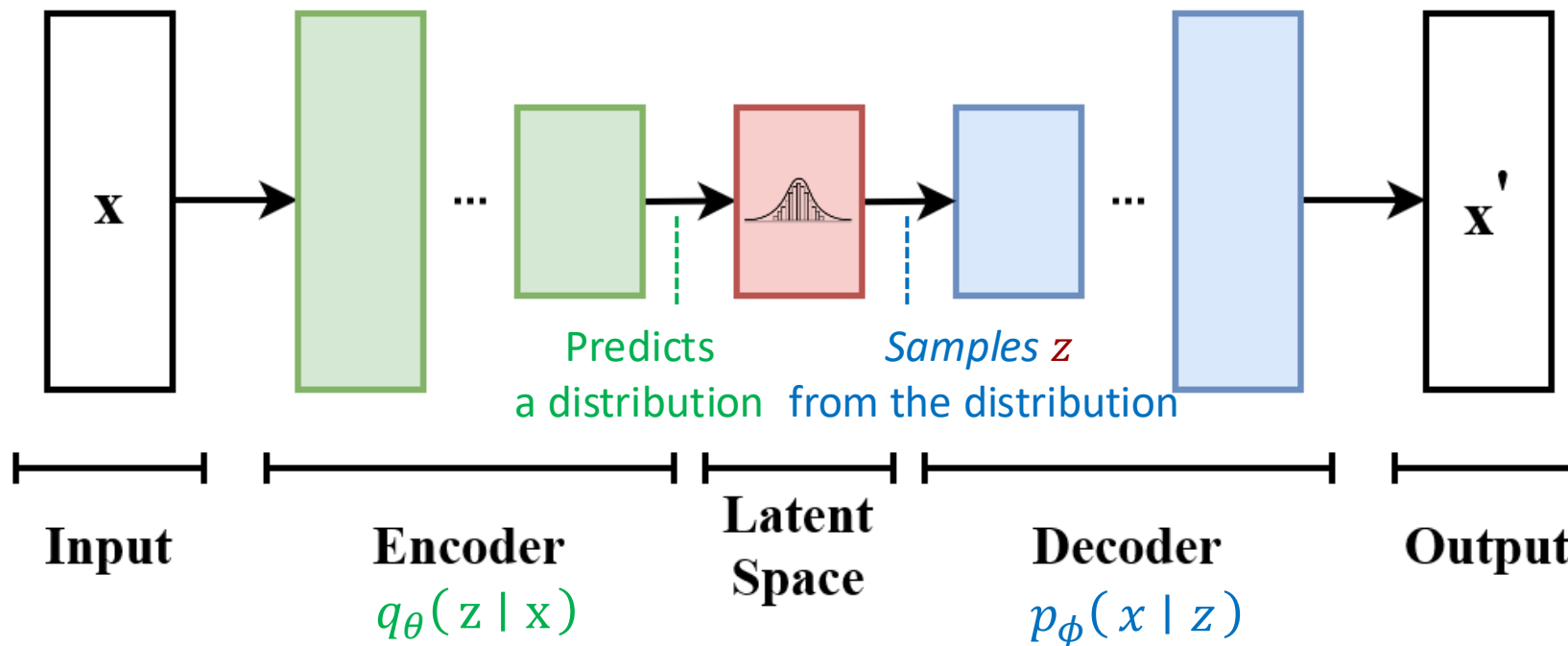


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*Kullback–Leibler (KL) divergence*



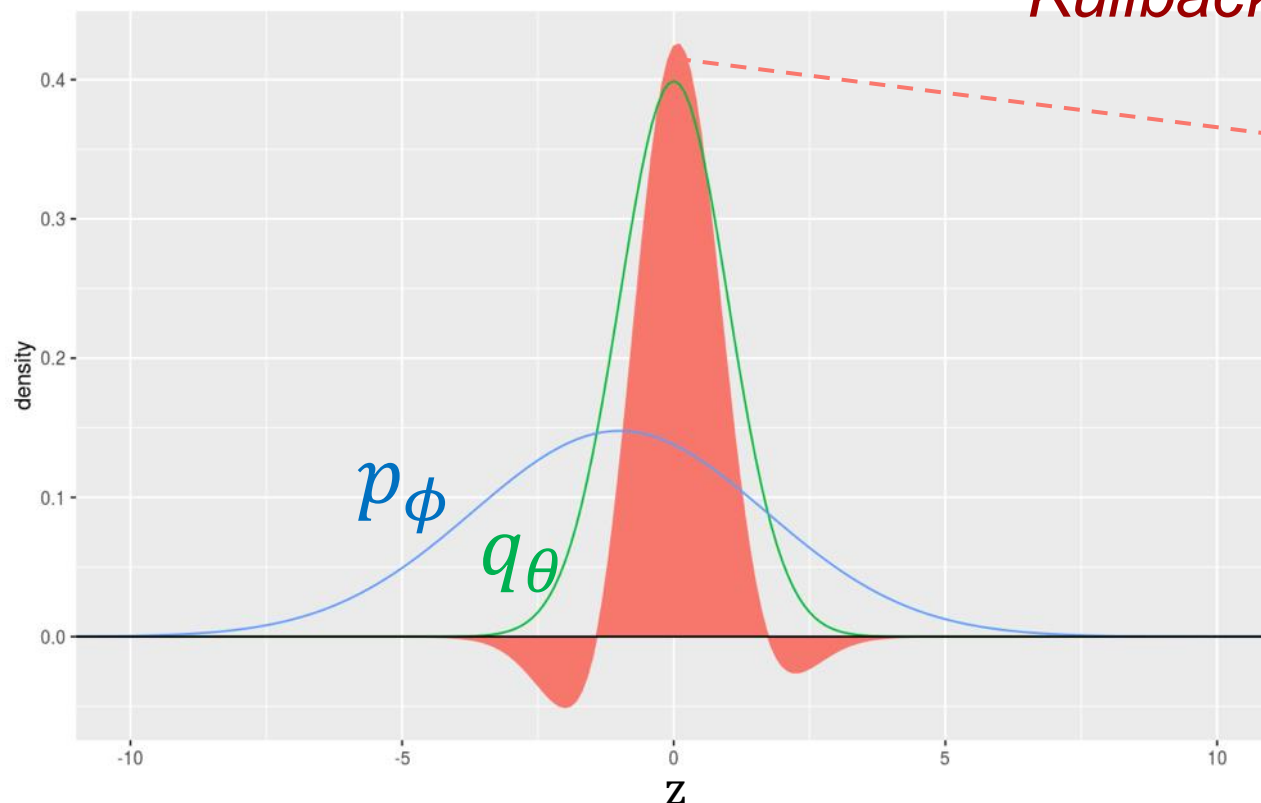


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**Kullback–Leibler (KL) divergence**



$$D_{KL}(q_{\theta}(\cdot | x) \parallel p_{\phi}(\cdot)) \\ := \int q_{\theta}(z | x) \log \frac{q_{\theta}(z | x)}{p_{\phi}(z)} dz$$



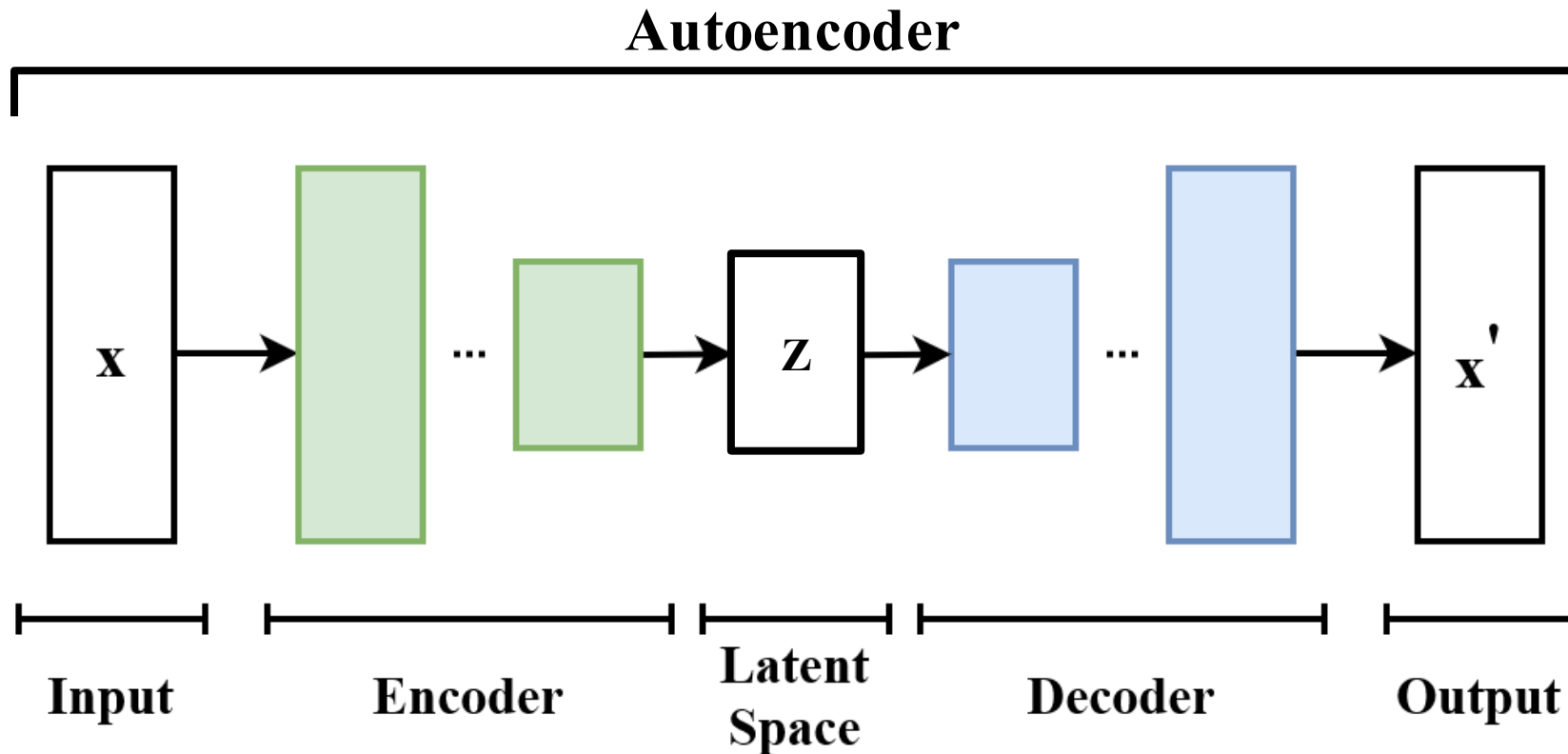
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# Generative Adversarial Networks (GANs)

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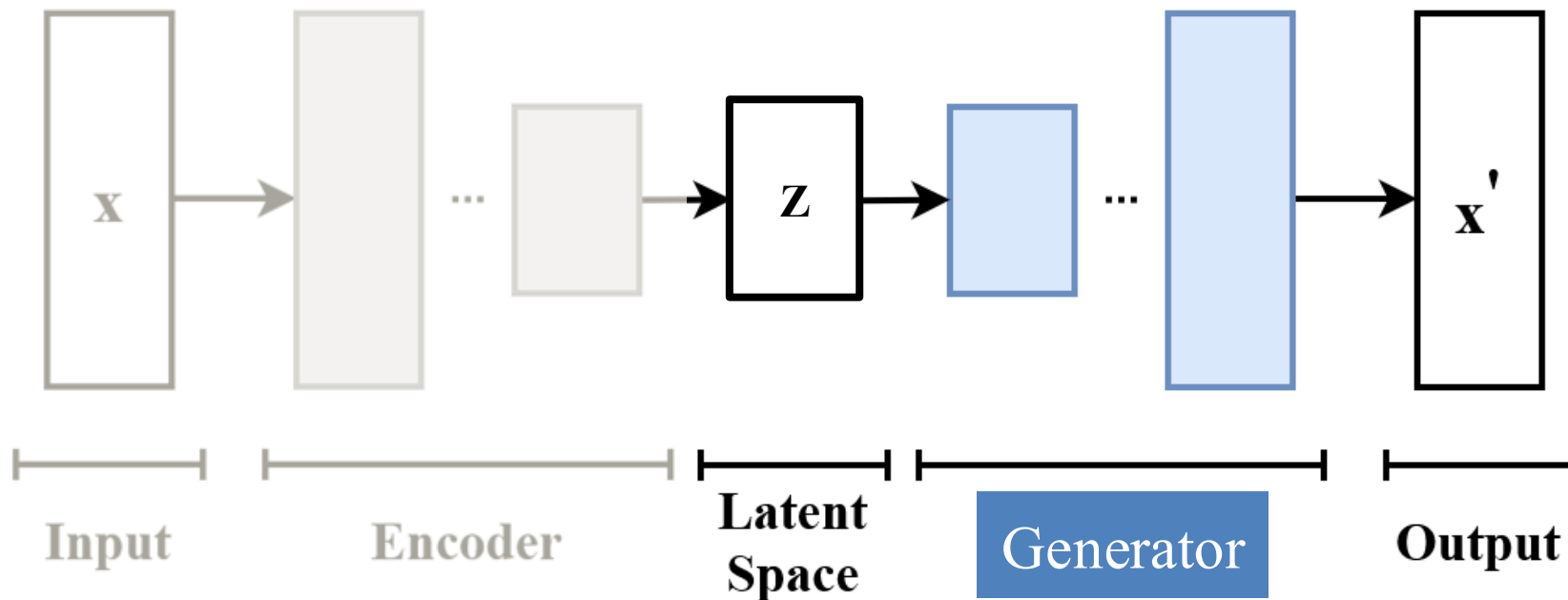
*Ian J. Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. "Generative adversarial nets." NeurIPS 2014.*

# From Autoencoders to GANs



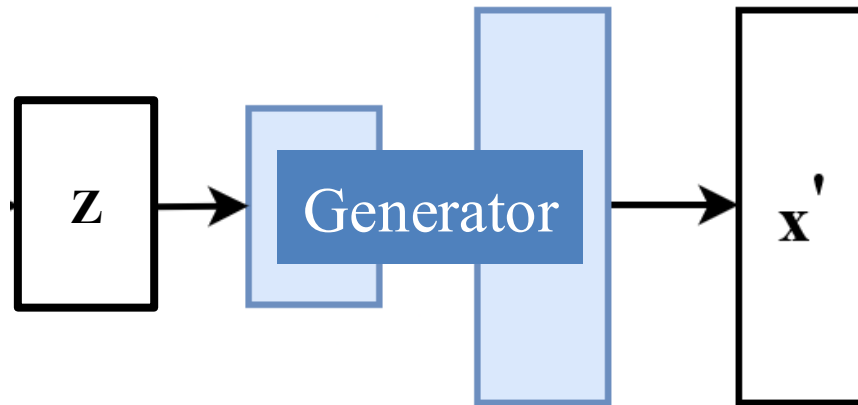
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- Goal: To generate a new  $x'$ .
- ➔ We only need the **Generator** !



# From Autoencoders to GANs

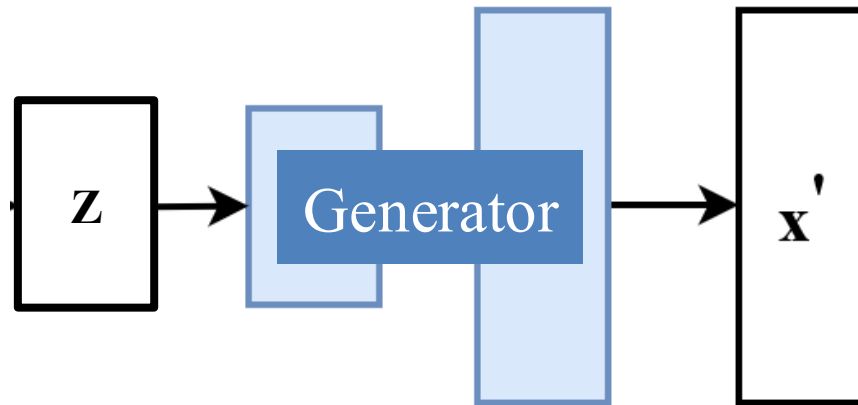
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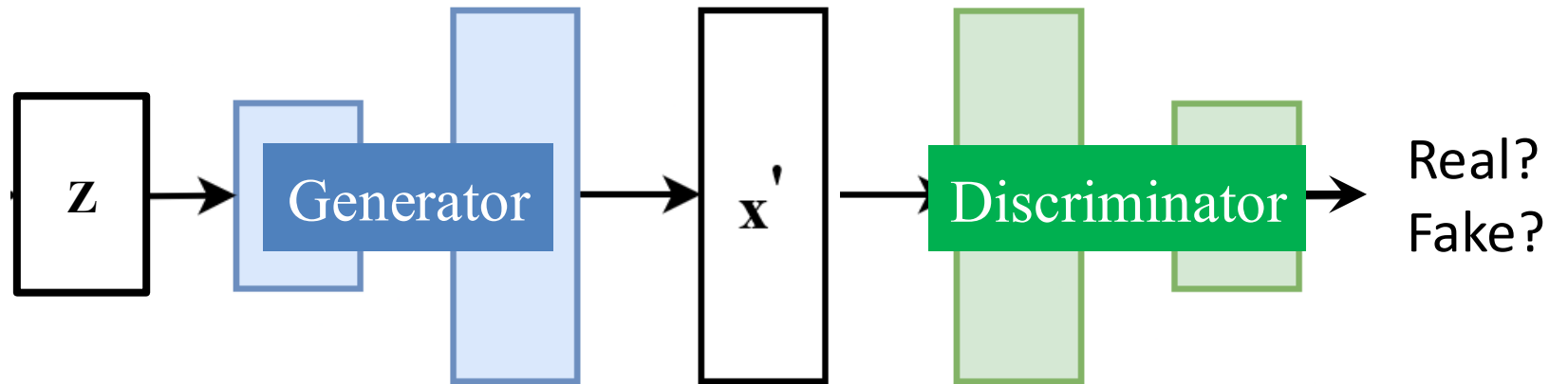
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- Goal: To generate a new  $x'$  that looks *real*.



# Training GANs: Two-Player Game

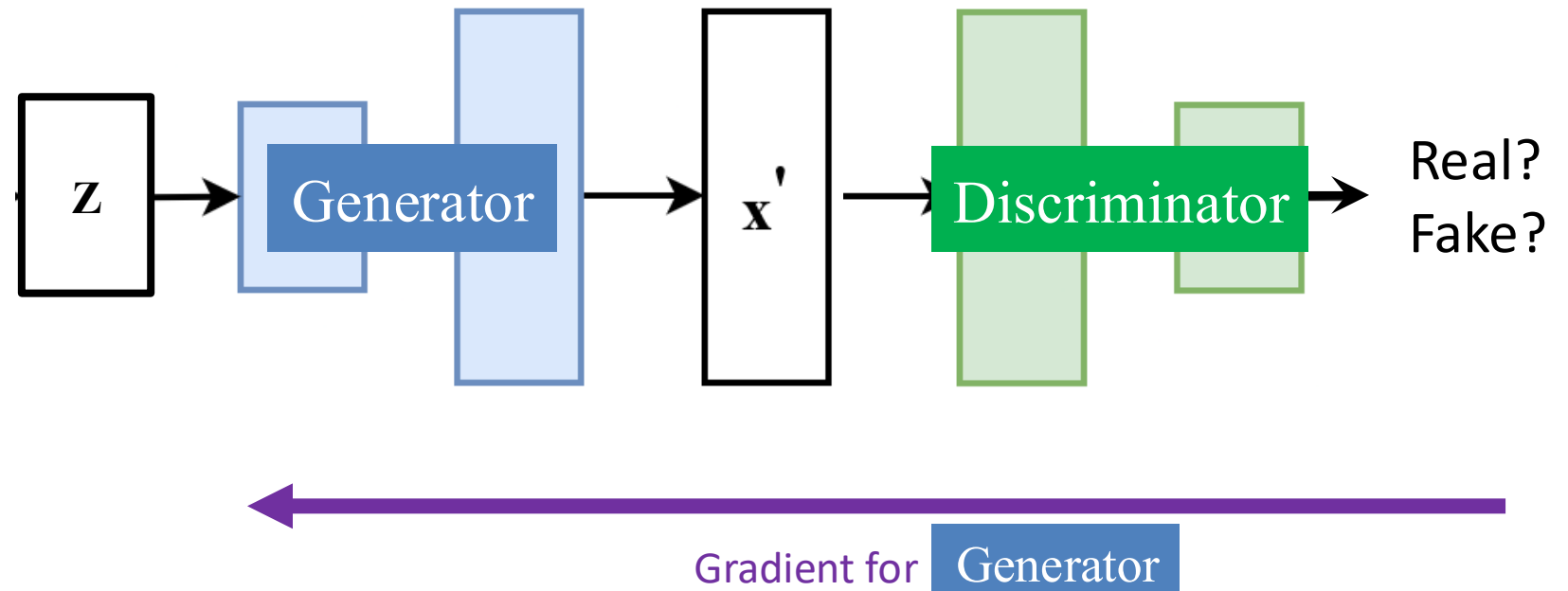
- Goal: To generate a new  $x'$  that looks *real*.
- ➔ Let's add a **Discriminator** which evaluates how *real*  $x'$  looks.  
(classifier)





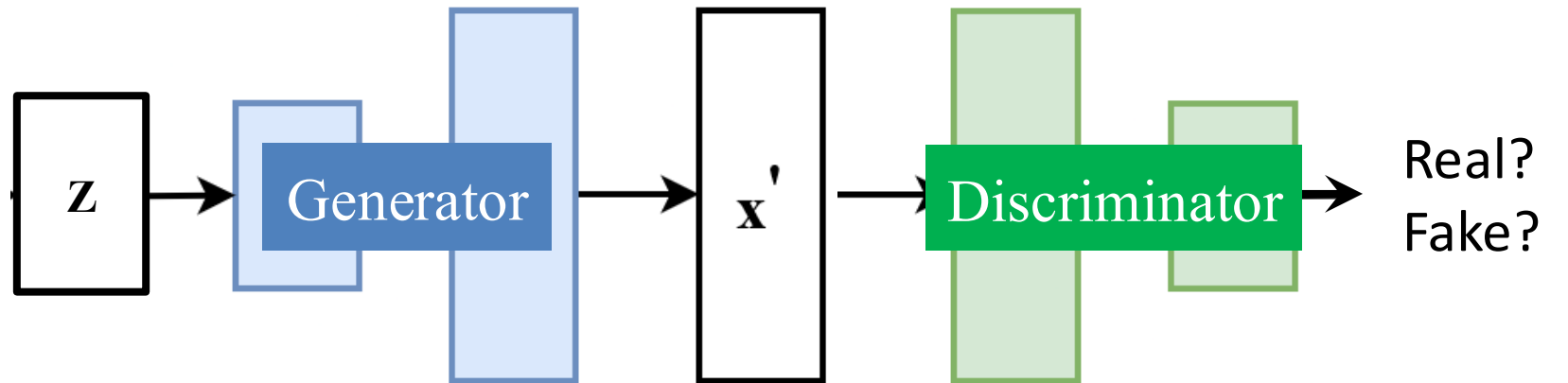
# Training GANs: Two-Player Game

- Goal: To generate a new  $x'$  that looks *real*.
- It's intuitive to learn the parameters of **Generator**.



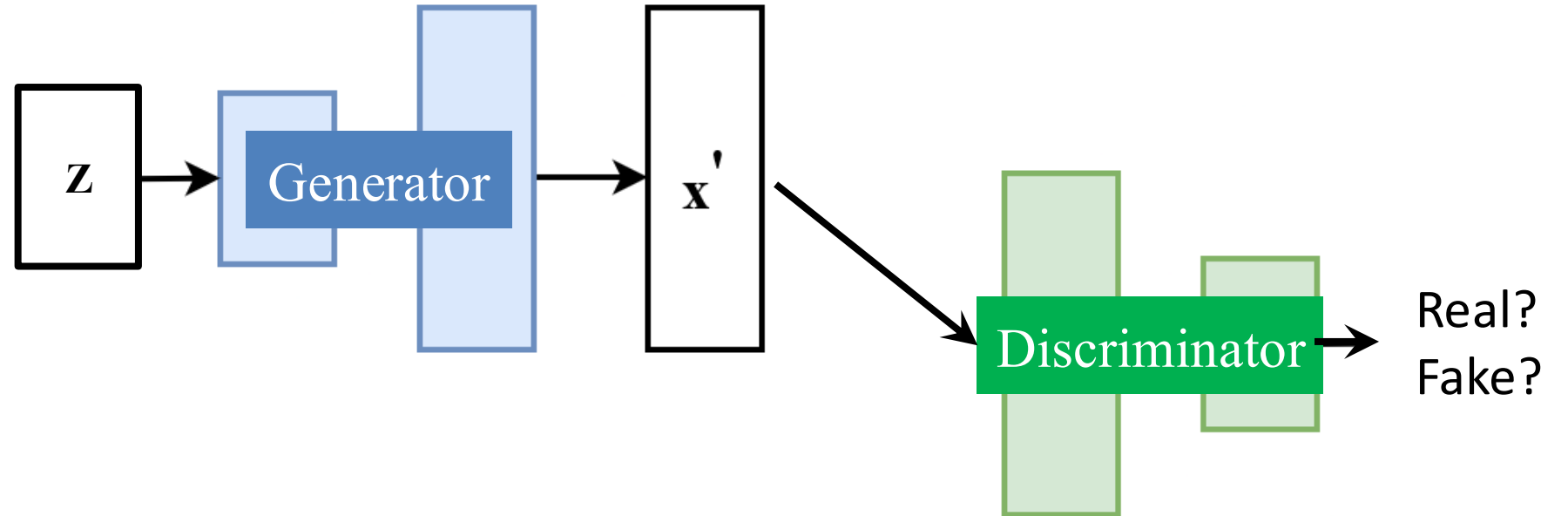
# Training GANs: Two-Player Game

- Goal: To learn the parameters of **Discriminator**.



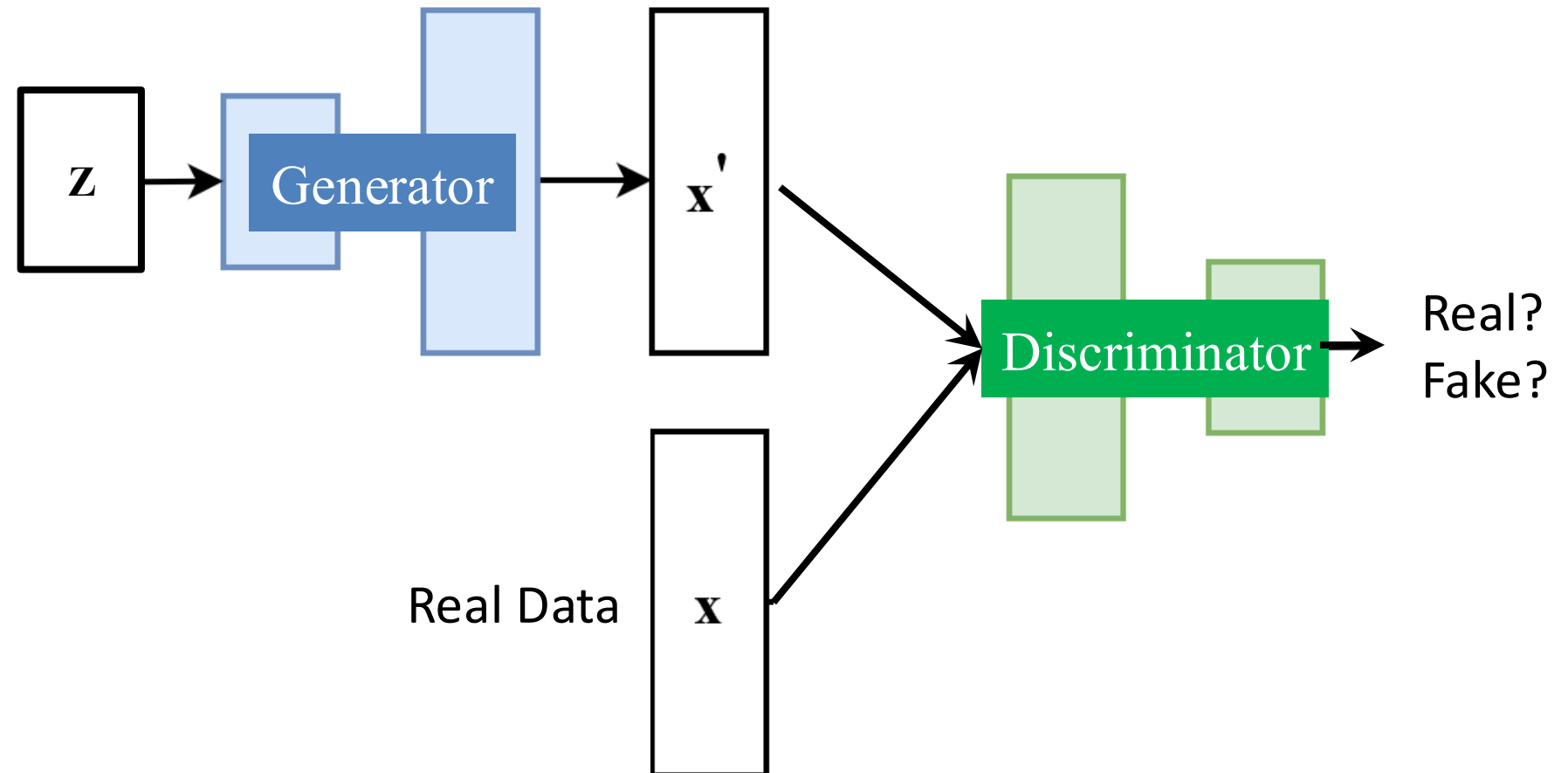
# Training GANs: Two-Player Game

- Goal: To learn the parameters of **Discriminator**.



# Training GANs: Two-Player Game

- Goal: To learn the parameters of **Discriminator** . ✓



# Summary & Key Takeaways

|                                |                                                                               |                                      |
|--------------------------------|-------------------------------------------------------------------------------|--------------------------------------|
| <b>CNNs</b>                    | Best suited for extracting spatial features from grid and image data          | e.g., raster data, satellite photos  |
| <b>GNNs</b>                    | Built to handle irregular topological data through message passing            | e.g., road networks                  |
| <b>RNNs &amp; Transformers</b> | The standard for modeling sequences and temporal dependencies                 | e.g., GPS trajectories               |
| <b>VAEs &amp; GANs</b>         | Powerful generative approaches for creating new, realistic data distributions | e.g. synthetic trajectory generation |